What does the pandemic surge in (U.S.) startups mean?

Ryan Decker, *Federal Reserve Board*

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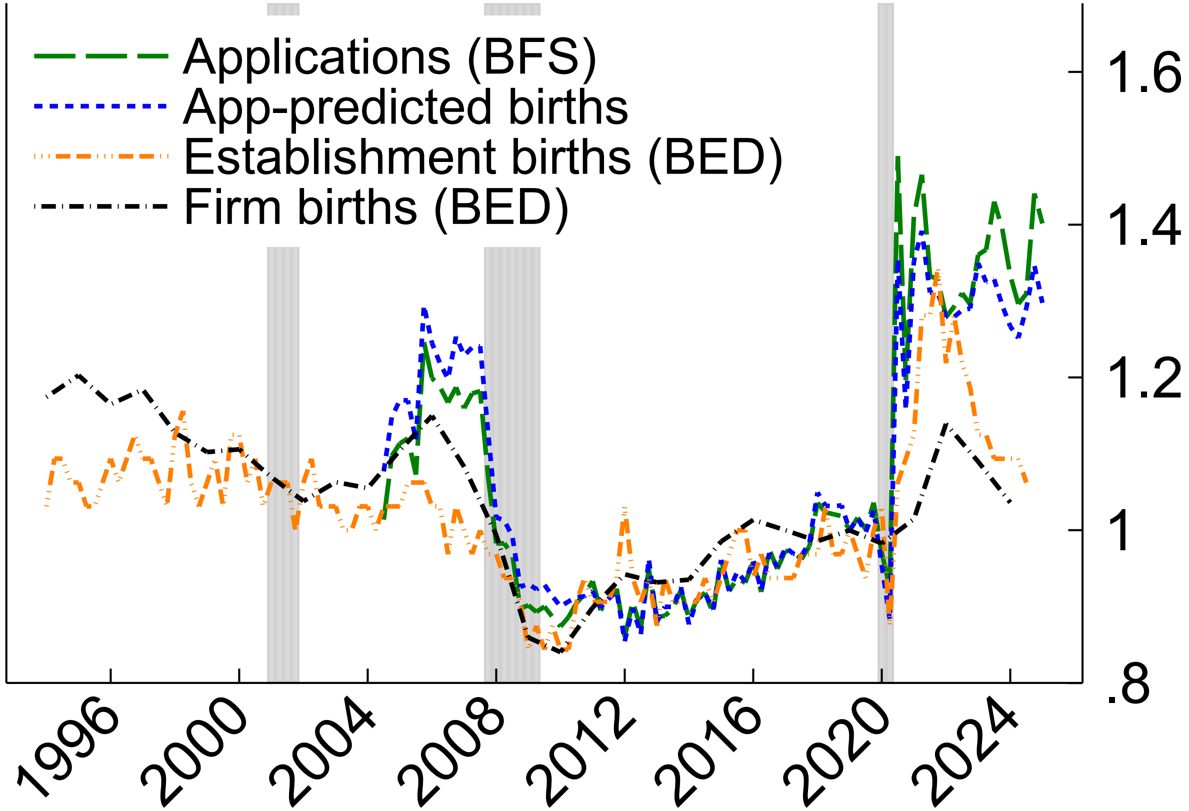
June 17, 2025

Without implication, this presentation is based largely on joint work with John Haltiwanger (U Maryland, NBER)

- "Surging business formation in the pandemic: Causes and consequences", Fall 2023 BPEA
- "High tech business entry in the pandemic era", FEDS Note
- "Surging business formation in the pandemic: A brief update", mimeo

The striking business entry surge in the U.S.

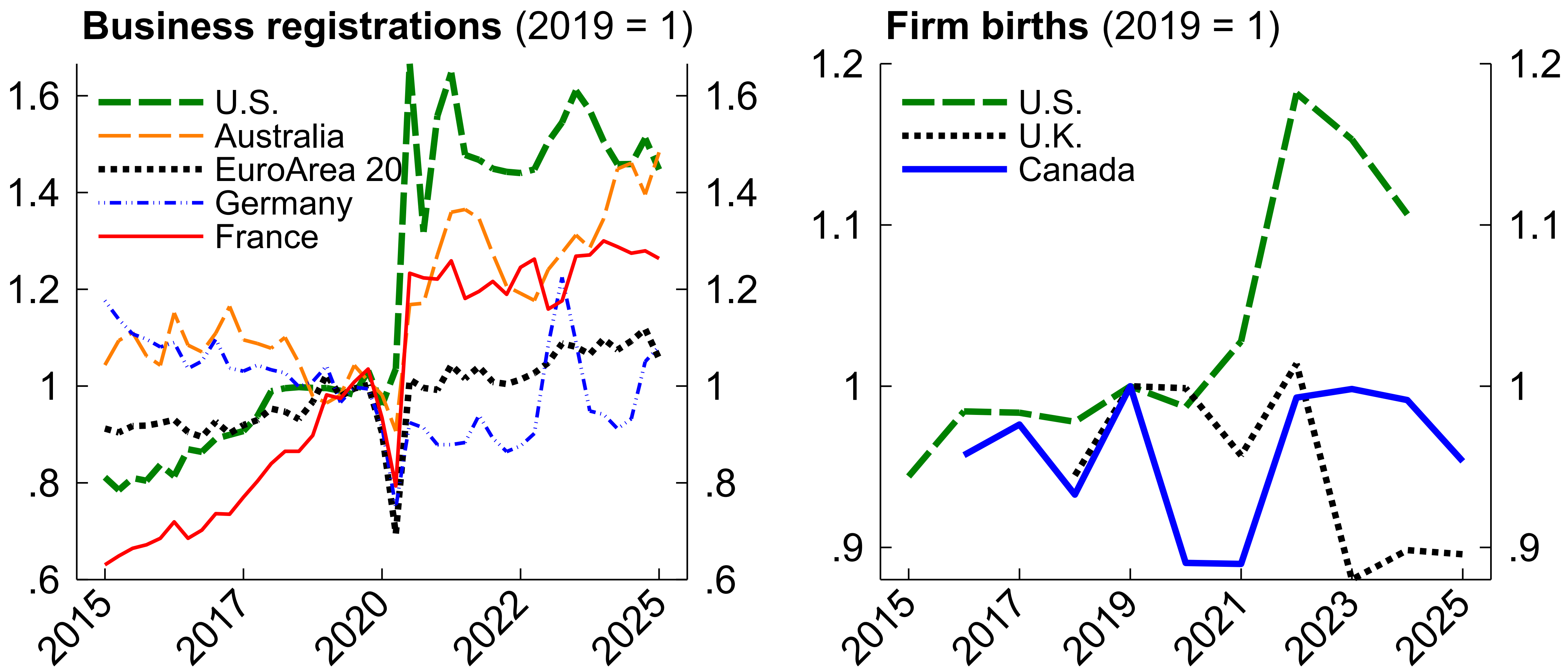
Entry rate indexes (2019:Q1 = 1)



Note: Applications are likely employer (HBA).
All series expressed as rates except BFS. 8-quarter prediction

- **Business applications**/registrations surged starting in mid-2020.
  - High-quality applications up 35%
  - **Predicted firm births** based on detailed application characteristics.
- Followed by surging ***Establishment* births:**
  - 1 million jobs per quarter, 2021:Q2-2024:Q3 (vs. 850k in 2019)
  - “Establishment” = business location
- And ***firm* births** jumped starting in year through March 2022
  - 1.9 million jobs per year, 2022-24 (highest since 2007)
  - “Firm” = company

U.S. surge is relatively strong



Note: U.S. registrations is total applications. Australia and U.K. seasonally adjusted by authors.
Canada and UK are 4-quarter trailing moving averages matched to Q1 reference.
Source: BFS, BED, Haver

What does the U.S. entry surge mean? Through the lens of *pandemic stories*

1. Geographic reallocation
2. Industry patterns
3. The policy environment (preliminary)
4. (If time) Labor market dynamics

1. Geographic reallocation (first few years)

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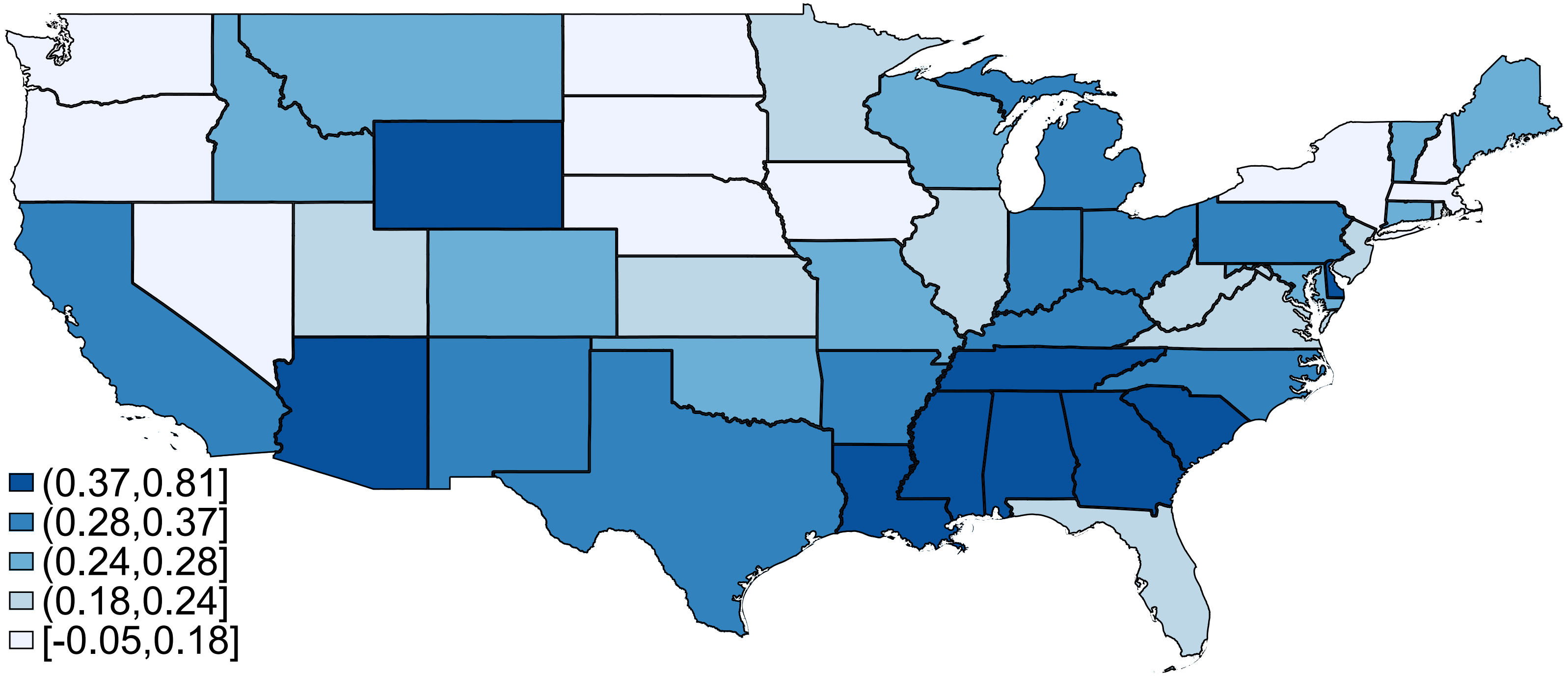
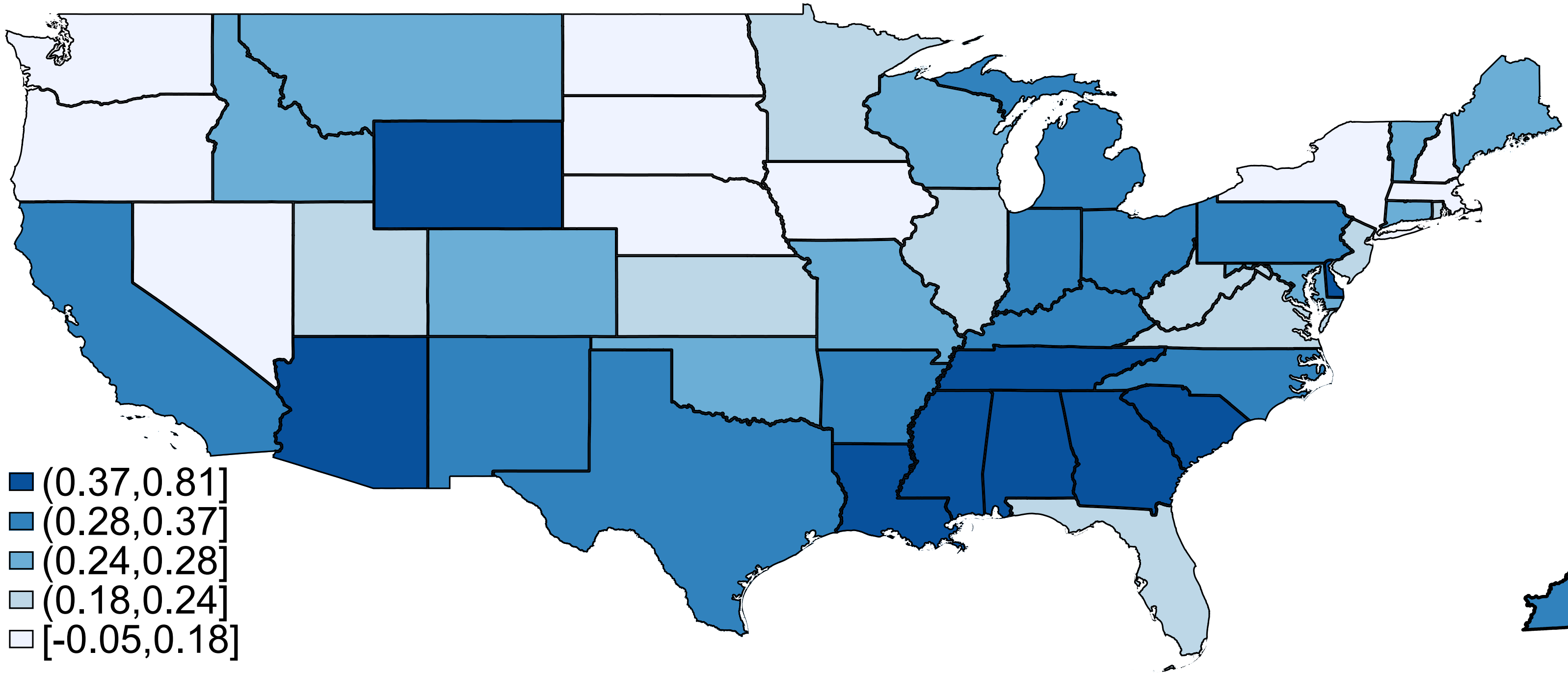


Note: Difference of average (log) likely employer applications per capita, 2020-2023 vs. 2010-2019.
Source: Census Bureau Business Formation Statistics and population estimates.

See also O’Brien 2022; Newman & O’Brien 2023; Newman & Fikri 2024

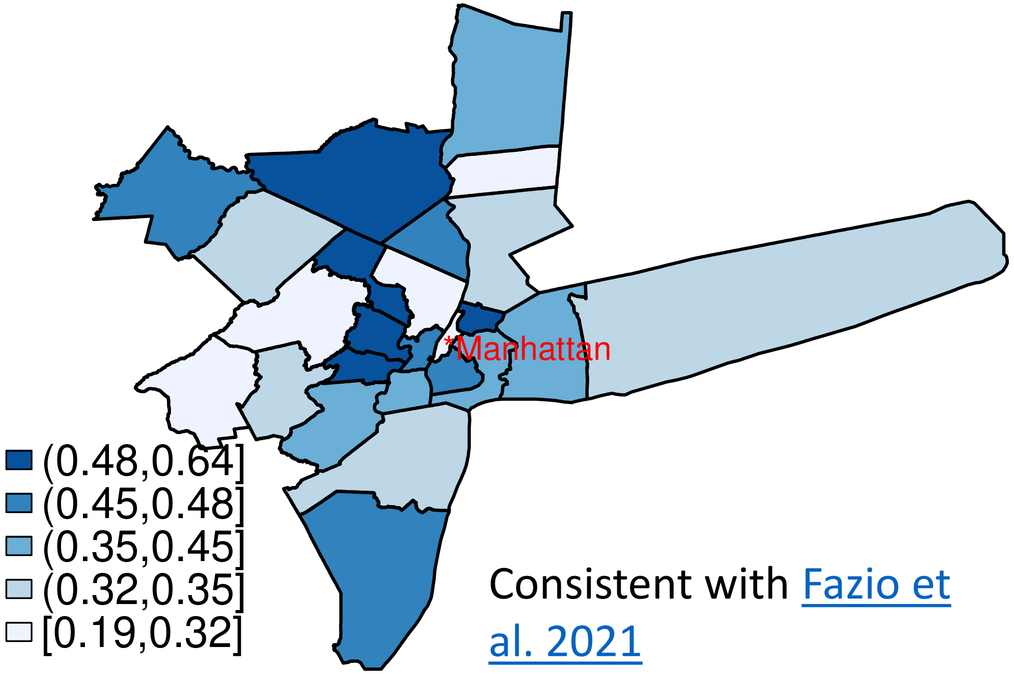
Note: State data for likely employer applications; county data for total applications

1. Geographic reallocation (first few years)



Note: Difference of average (log) likely employer applications per capita, 2020-2023 vs. 2010-2019.
Source: Census Bureau Business Formation Statistics and population estimates.

- **Donut effect** in cities related non-linearly to pop density, estab density, and changes in WFH.



Consistent with [Fazio et al. 2021]

See also O'Brien 2022; Newman & O'Brien 2023; Newman & Fikri 2024

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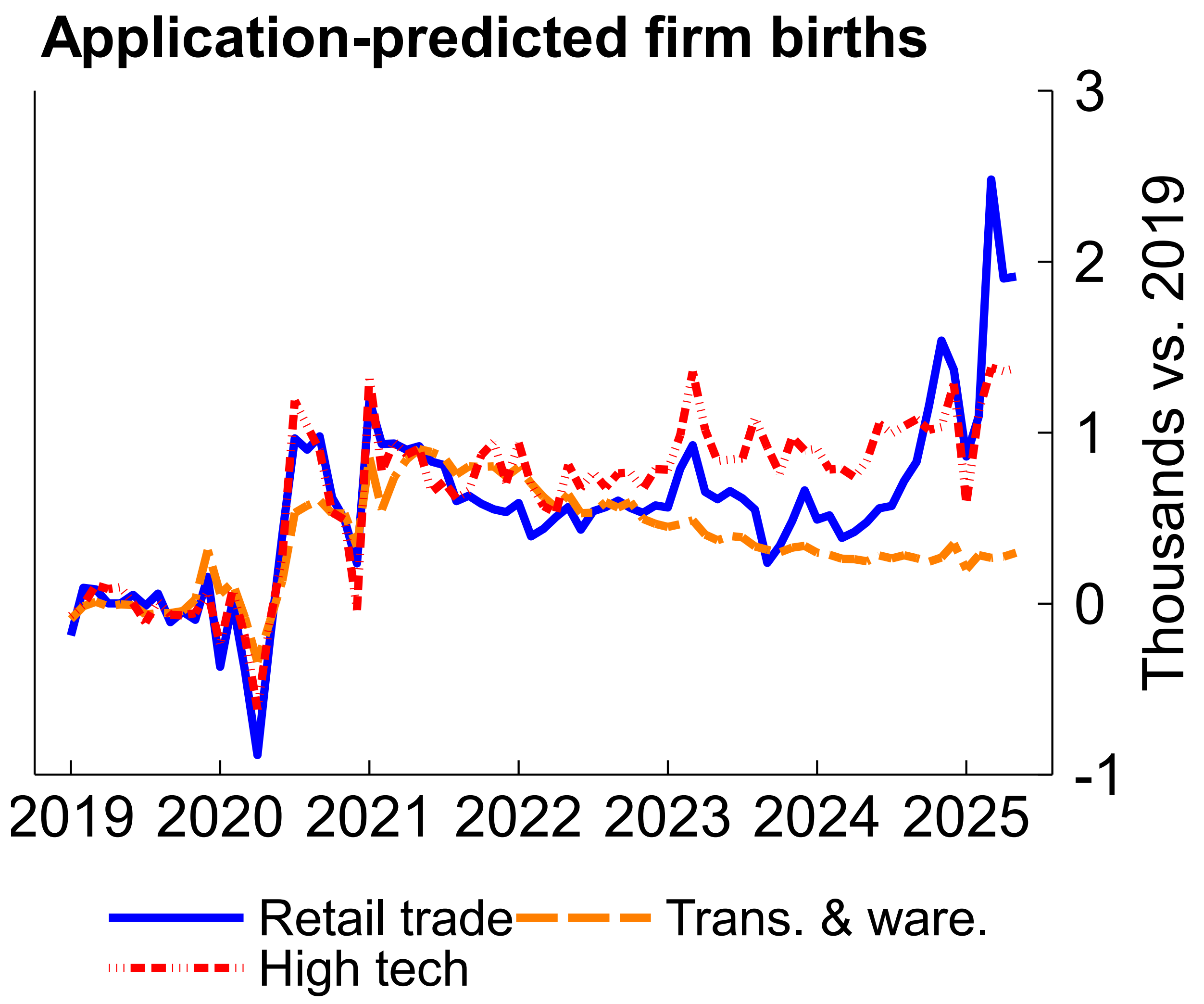
2. Industry patterns

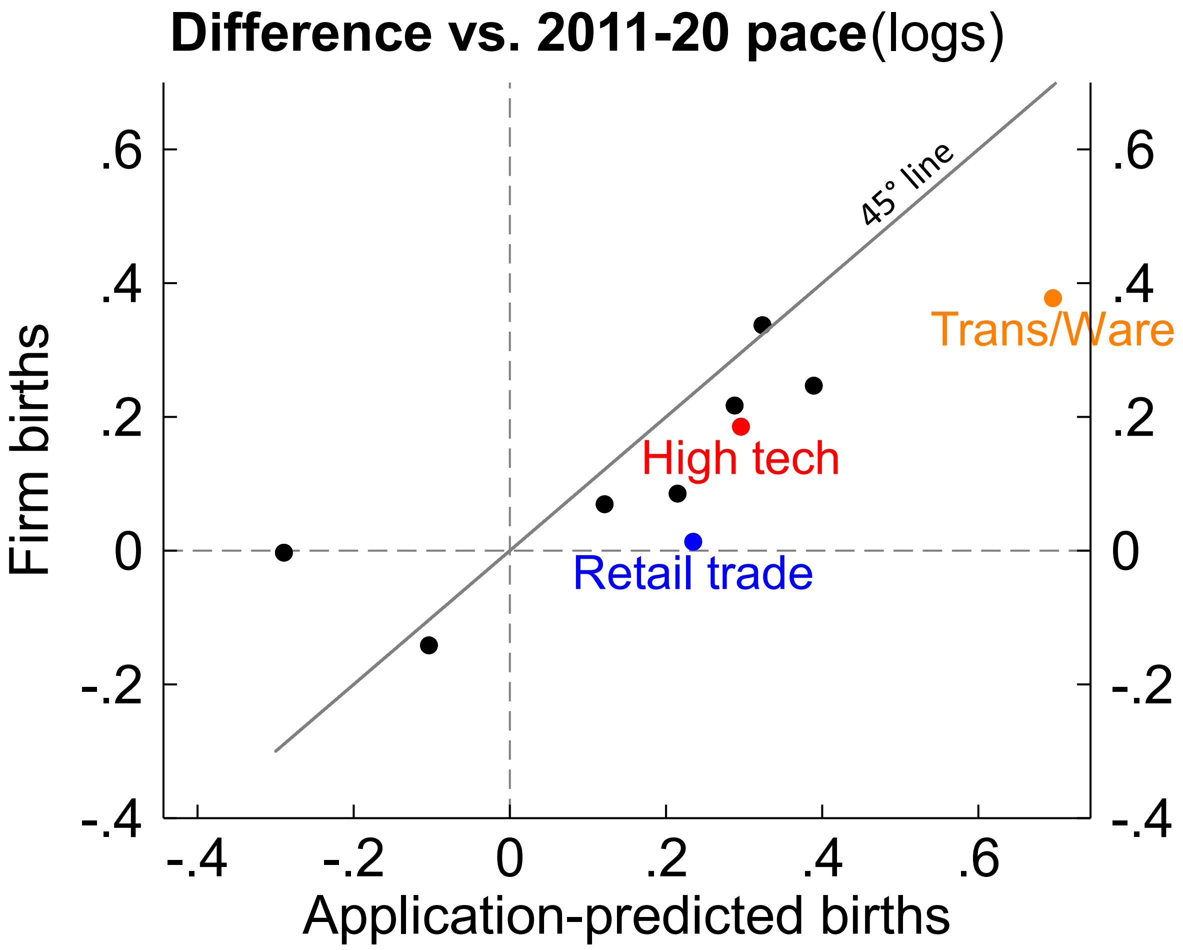
2. Industry patterns: Applications in retail, logistics, tech



Note: "High tech" here combines NAICS 51 and 54

- **Retail trade:**
  - Strong initial surge, some cooling, then recent resurgence
  - Driven largely by **online retail**
  - (Next slide) Low transition rates
- **Transportation and warehousing:**
  - Surge peaks during 2021 supply chain crises
  - e.g., nonscheduled air transport; couriers & delivery; truck transportation; freight arrangement; warehousing & storage; transportation support
- **High tech**
  - Steady surge, gradual gains recently

2. Industry patterns: Application -> prediction -> births

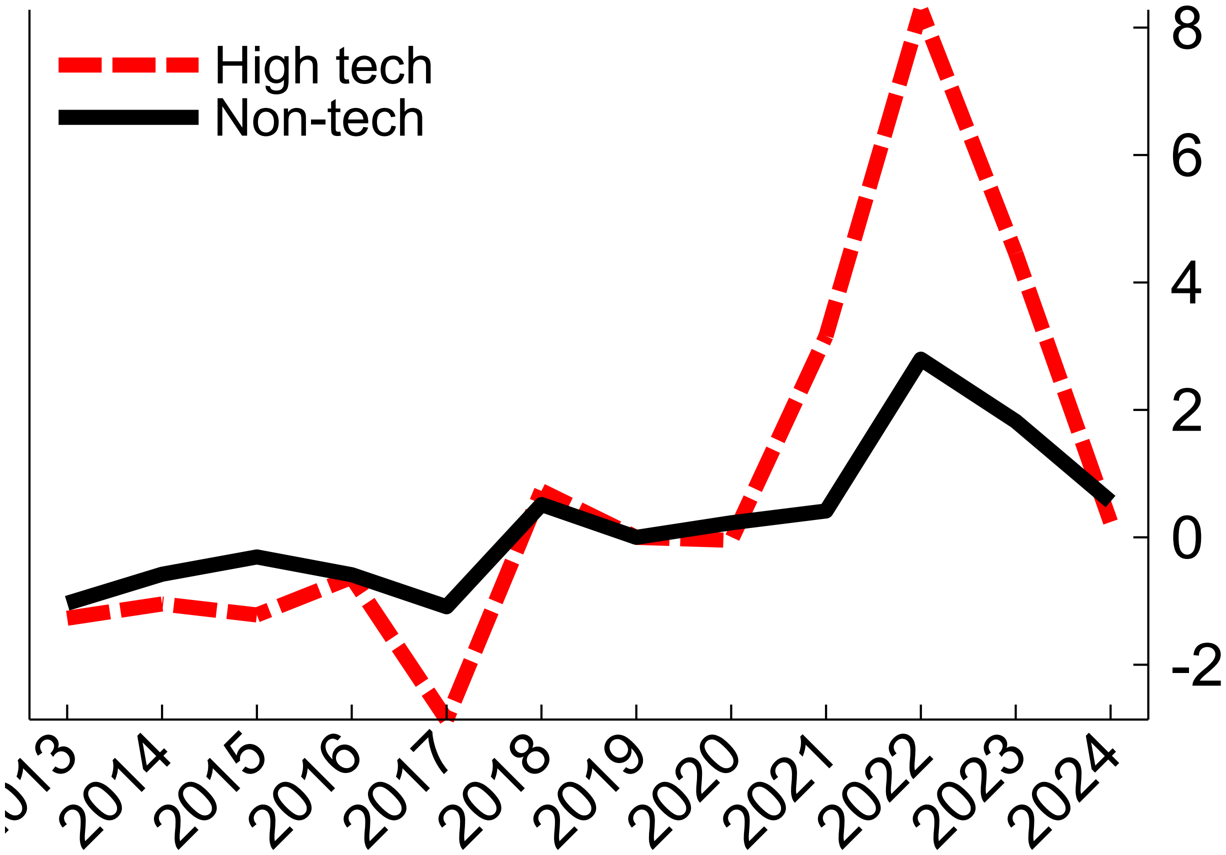


- Not all applications transition to firm births
- Census Bureau's application-based "predicted firm births" did well in most sectors…
- …But not **retail trade**. Likely a combination of
  - "Cheap talk" applications
  - Nonemployer merchant activity

Note: "High tech" here combines NAICS 51 and 54

2. Industry patterns: Surging high tech industries

Net establishment gains (logs; 2019 = 0)



Note: Annual (log) gains versus 2019. BDS tech definition.
Source: QCEW.

Surge industries include

- Software publishers
- Data processing, hosting
- Magnetic & optical media mfg
- Scientific R&D
- Pharma & medicine mfg
- Computer systems design
- Mgmt, sci, & tech consulting

Note: Some disagreement among U.S. data sources on the nature of the tech surge; more work needed

3. The (U.S.) policy environment (preliminary)

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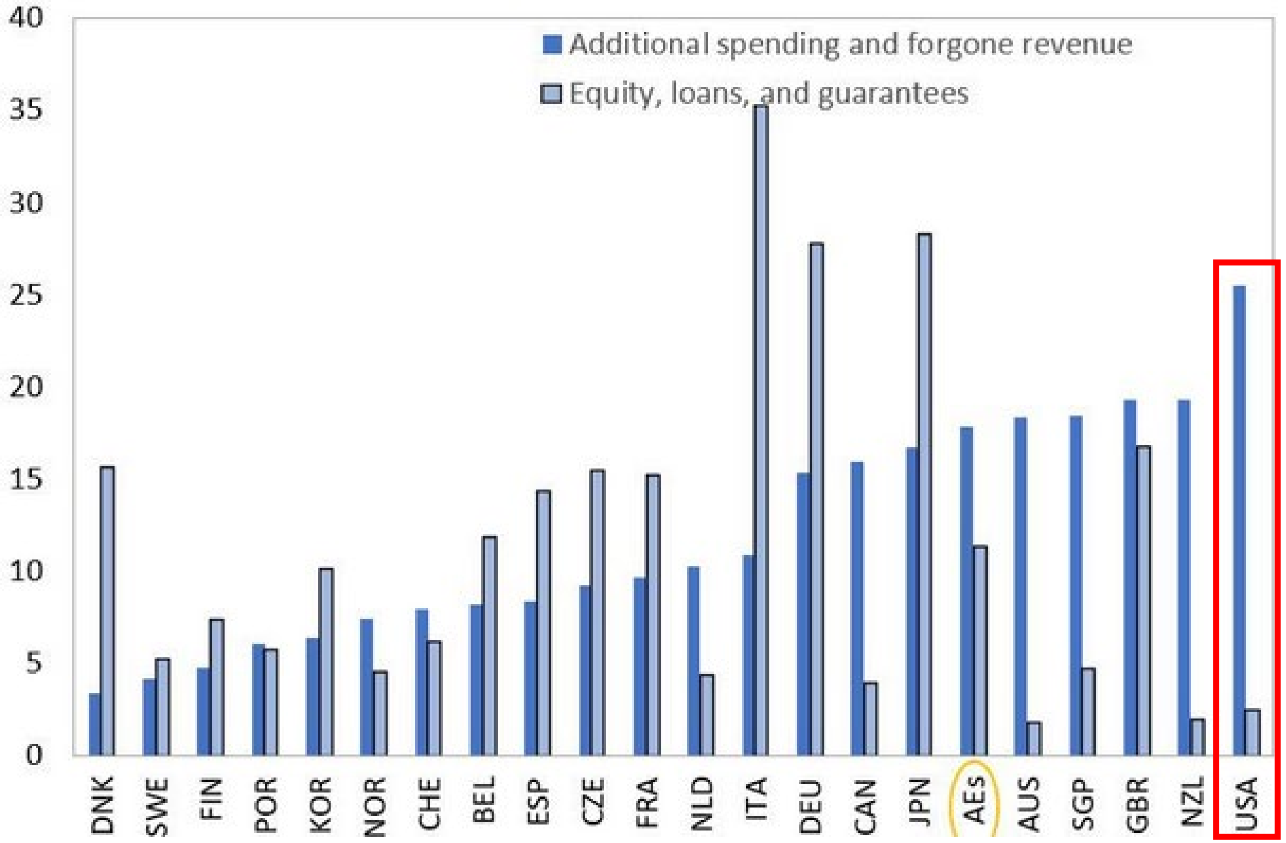
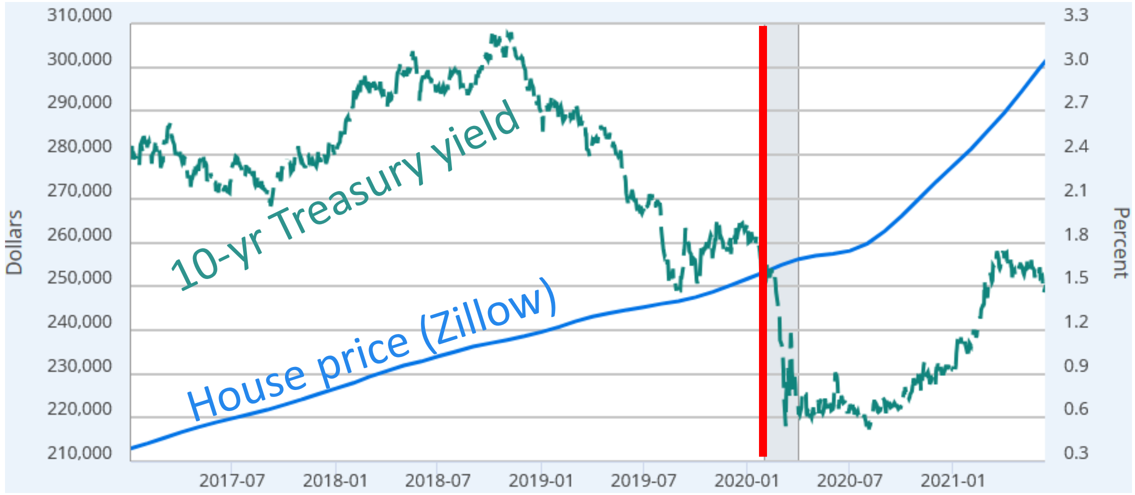
- Did the U.S. pandemic policy response induce entry?
  - Fiscal: cash to households, expanded unemployment insurance, direct support to businesses
  - Monetary policy easing → eased financial conditions





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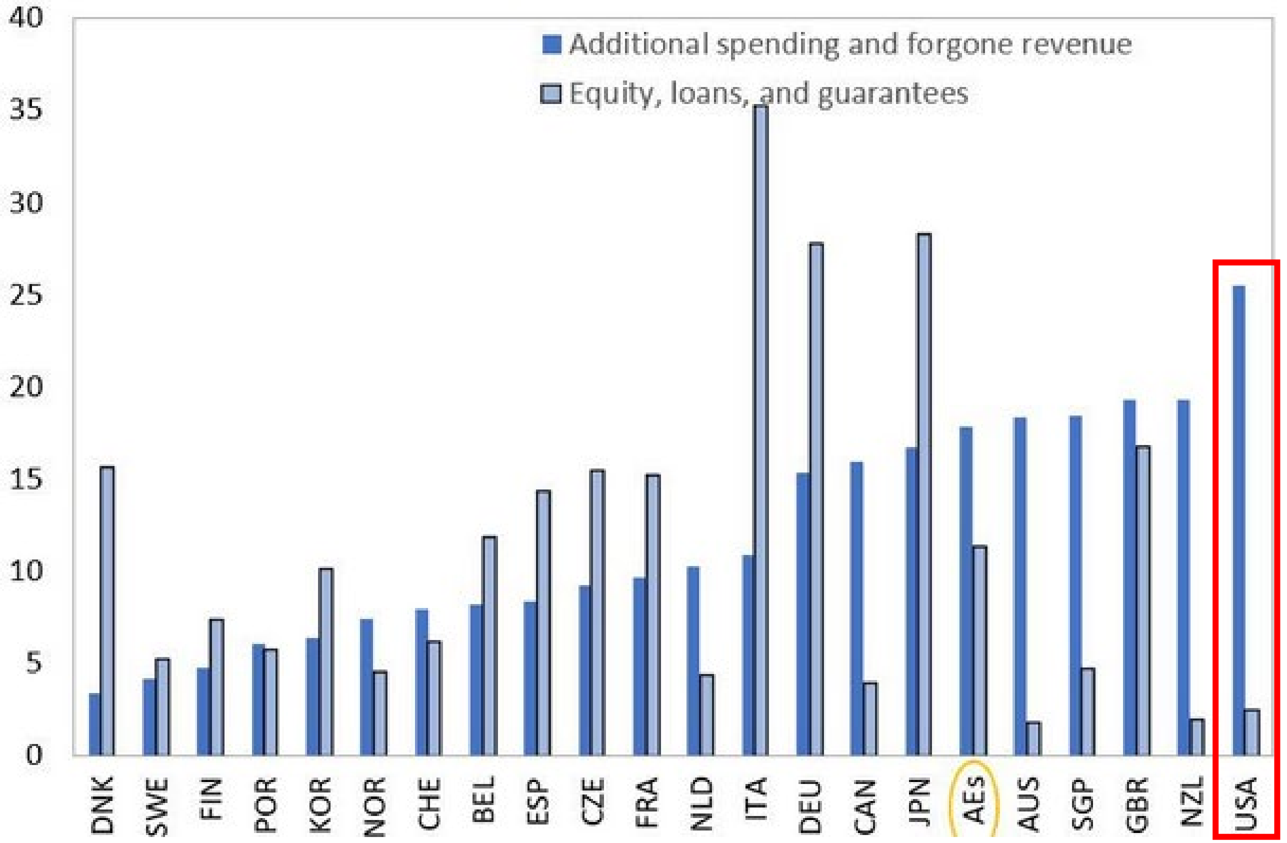
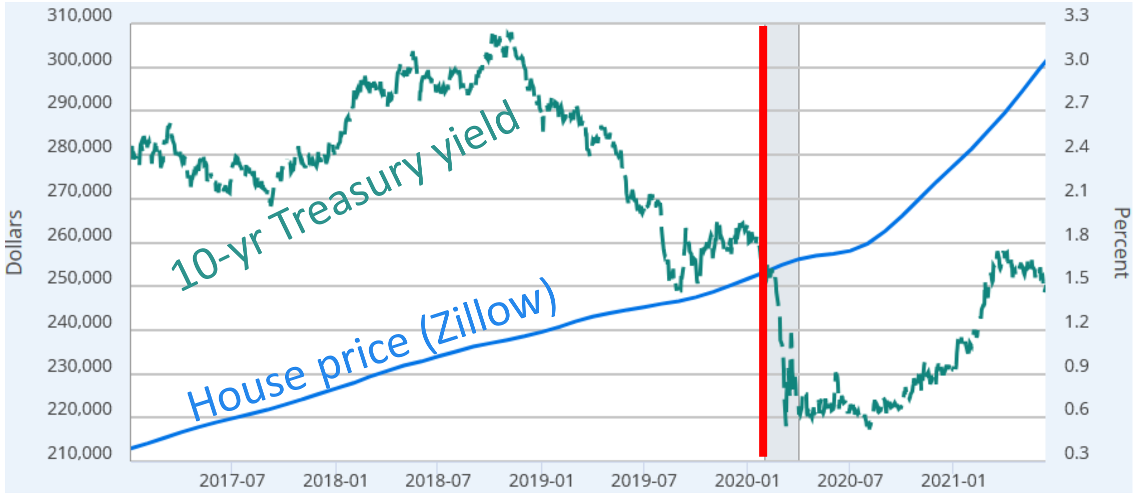
- Did low regulatory burden facilitate entry?





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- Did the U.S. pandemic policy response induce entry?
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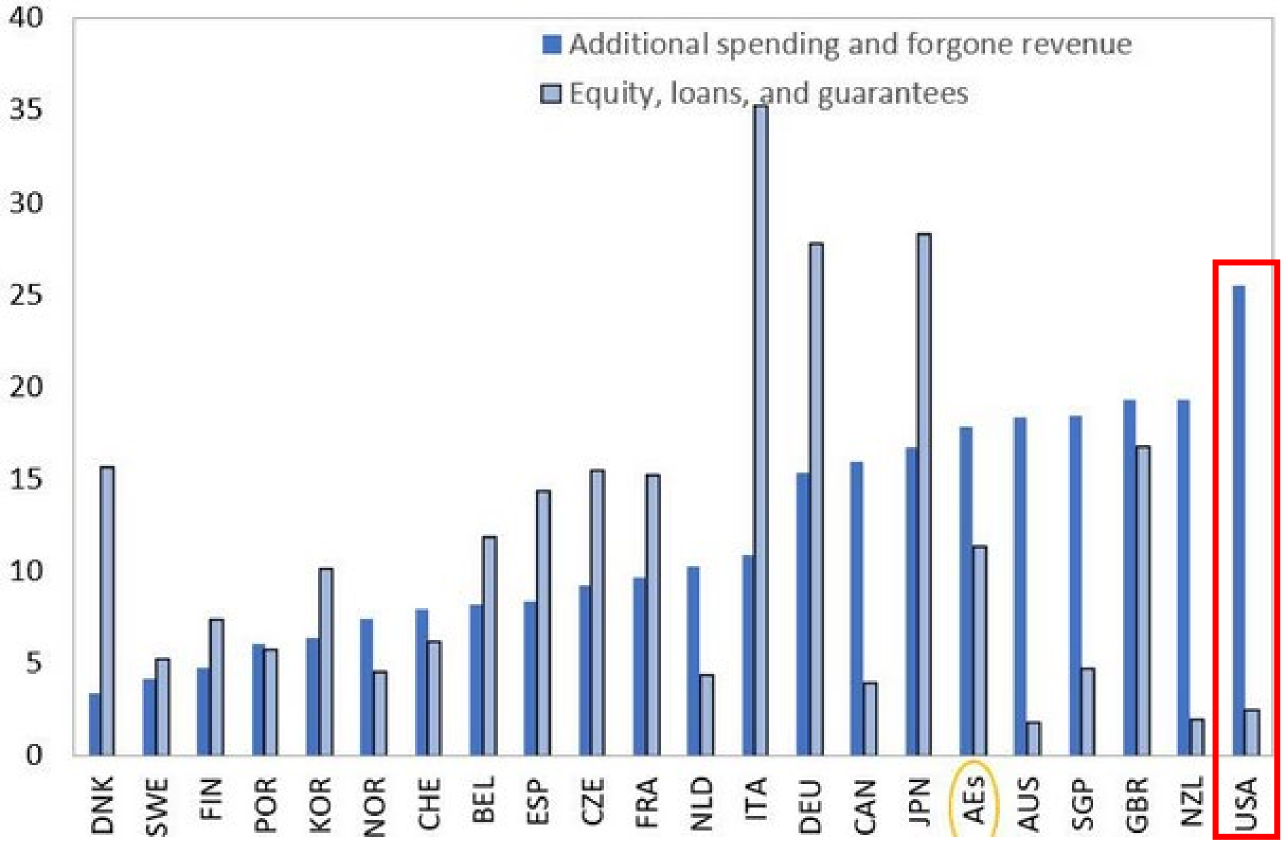
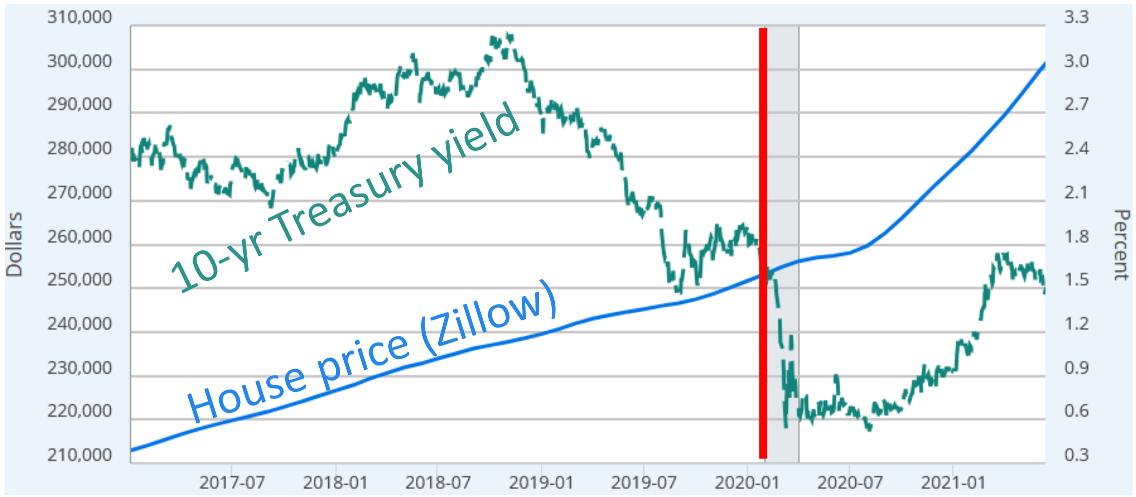
- Did low regulatory burden facilitate entry?
- For today: Use (within-U.S.) industry variation in historical policy exposure
  - For later? Use cross-country variation!





Industry fiscal beta

- **Fiscal beta:** Historical (pre-pandemic) correlation between fiscal stimulus and *industry* firm entry
    - Use discretionary fiscal effect from Cashin et al. 2018 (updated).

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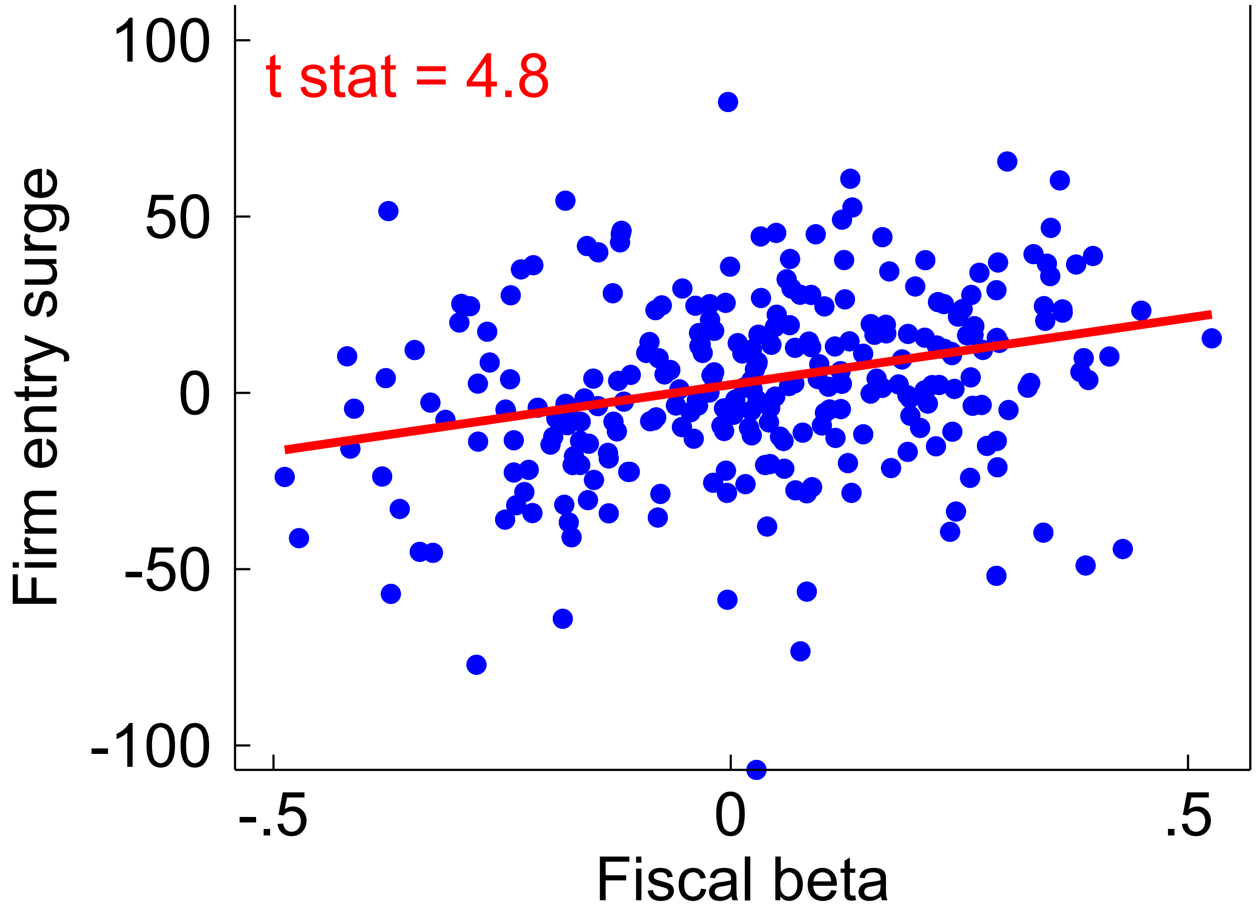
  - Use discretionary fiscal effect from [Cashin et al. 2018](#) (updated).

- **Result:** Fiscally sensitive industries saw larger entry surge
  - Consistent with [Fazio et al. (2021)](#); [Choi et al. (2024)](#)



Note: Entry surge in logs, 2021-22 vs. 2013-2019.
Regression line weighted by firm count.
Source: BDS, Cashin et al. (2018), author calculations.

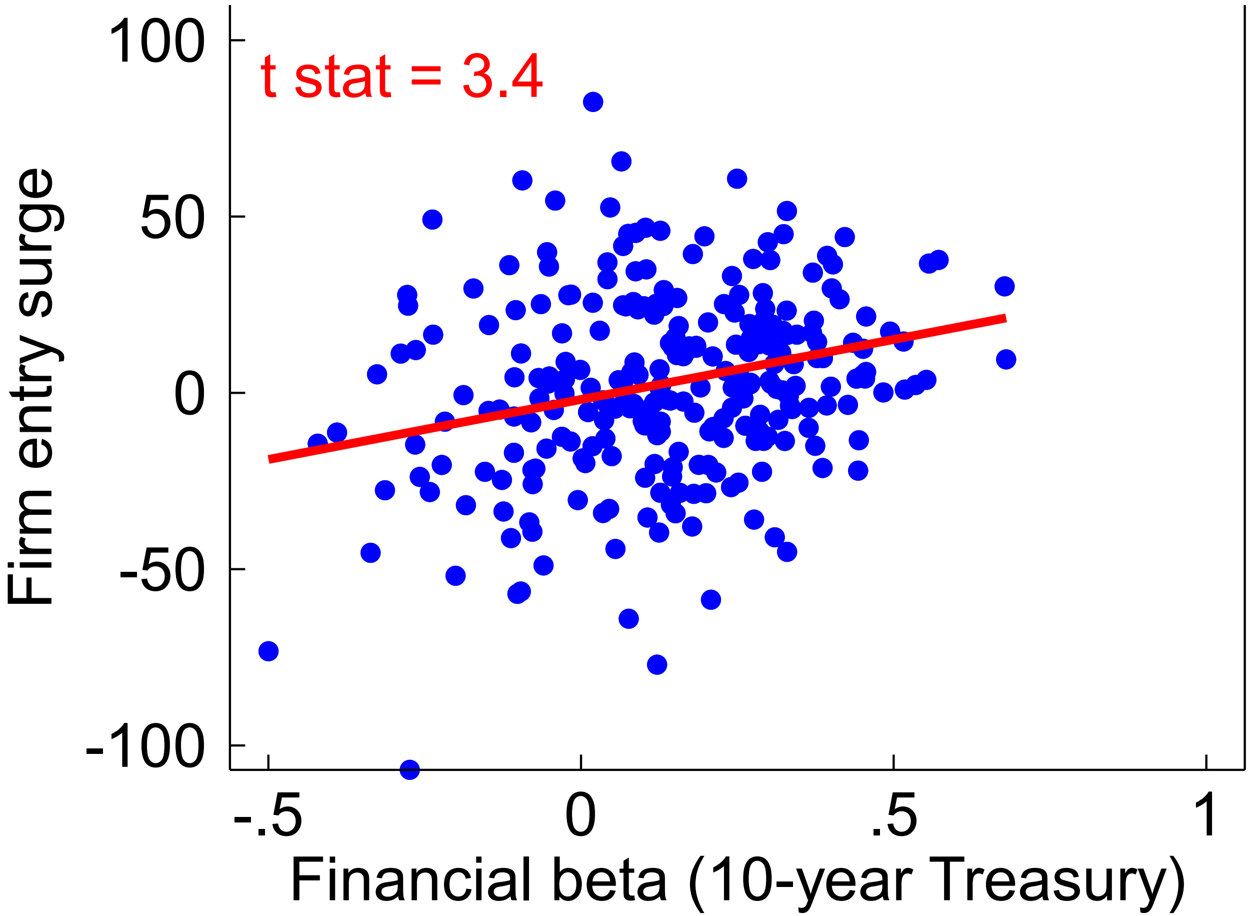
Industry financial beta - 10-year Treasury yield

- **10-year Treasury beta:** Historical (pre-pandemic) correlation between financial condition-based stimulus via 10-yr and *industry* firm entry
  - Use FCI-G from Ajello et al. 2023

Industry financial beta - 10-year Treasury yield

- **10-year Treasury beta:** Historical (pre-pandemic) correlation between financial condition-based stimulus via 10-yr and *industry* firm entry
  - Use FCI-G from Ajello et al. 2023

- **Result:** Industries sensitive to the 10-year saw larger entry surge
  - Consistent with Siemer 2019; Mehrotra & Sergeyev 2021



Note: Entry surge in logs, 2021-22 vs. 2013-2019.
Regression line weighted by firm count.
Source: BDS, Ajello et al. (2023), author calculations.

Industry financial beta – House prices

- **House price beta:** Historical (pre-pandemic) correlation between financial condition-based stimulus via house prices and *industry* firm entry
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Industry financial beta – House prices

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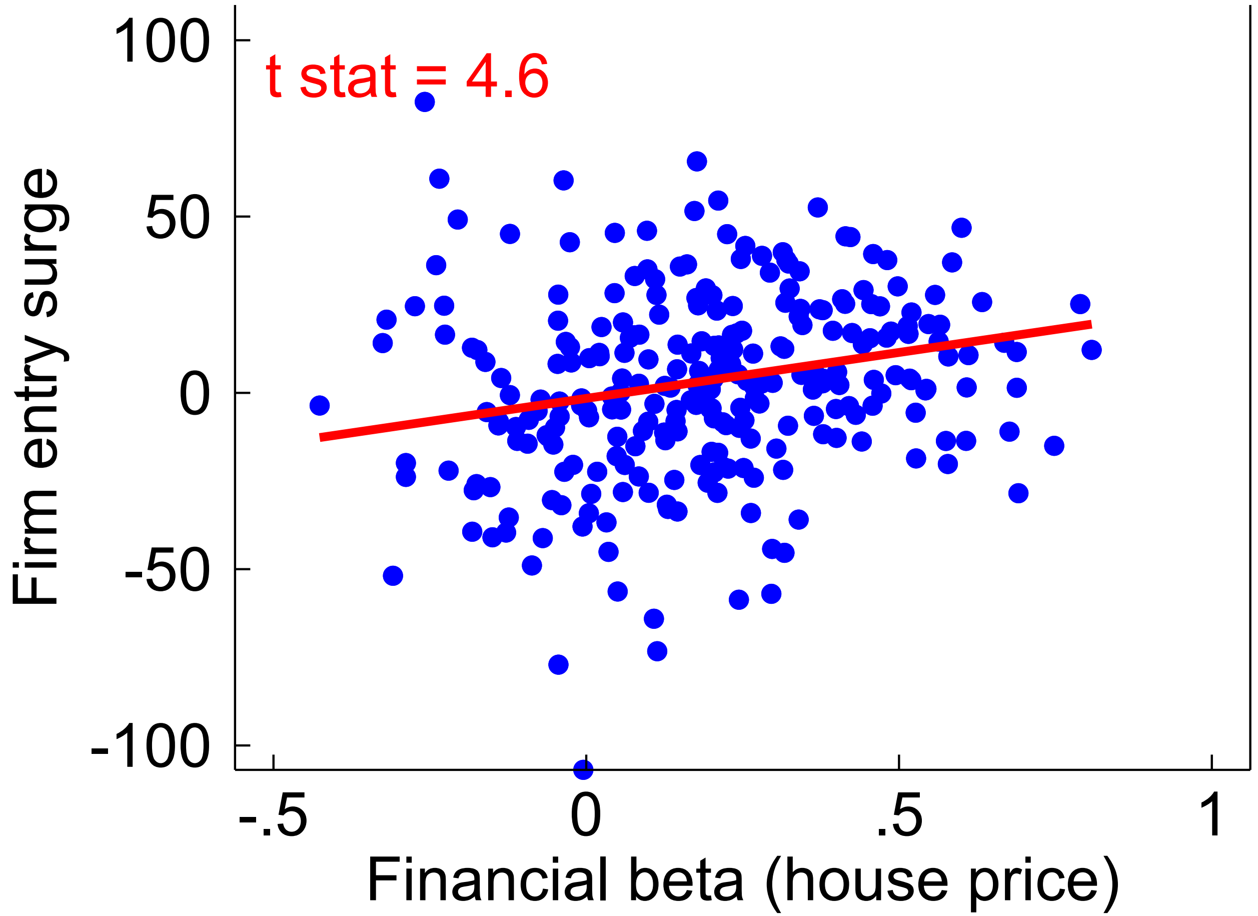
  - Use FCI-G from Ajello et al. 2023

- **Result:** Industries sensitive to the house prices saw larger entry surge
  - Consistent with Blackwood 2025; Davis & Haltiwanger 2024; Fort et al. 2013; Decker 2015.



Note: Entry surge in logs, 2021-22 vs. 2013-2019.
Regression line weighted by firm count.
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Industry regulation

- **Regulation:** Industry-level federal regulatory burden as of 2019
  - Use Mercatus RegData 4.1 from Al-Ubaydli & McLaughlin

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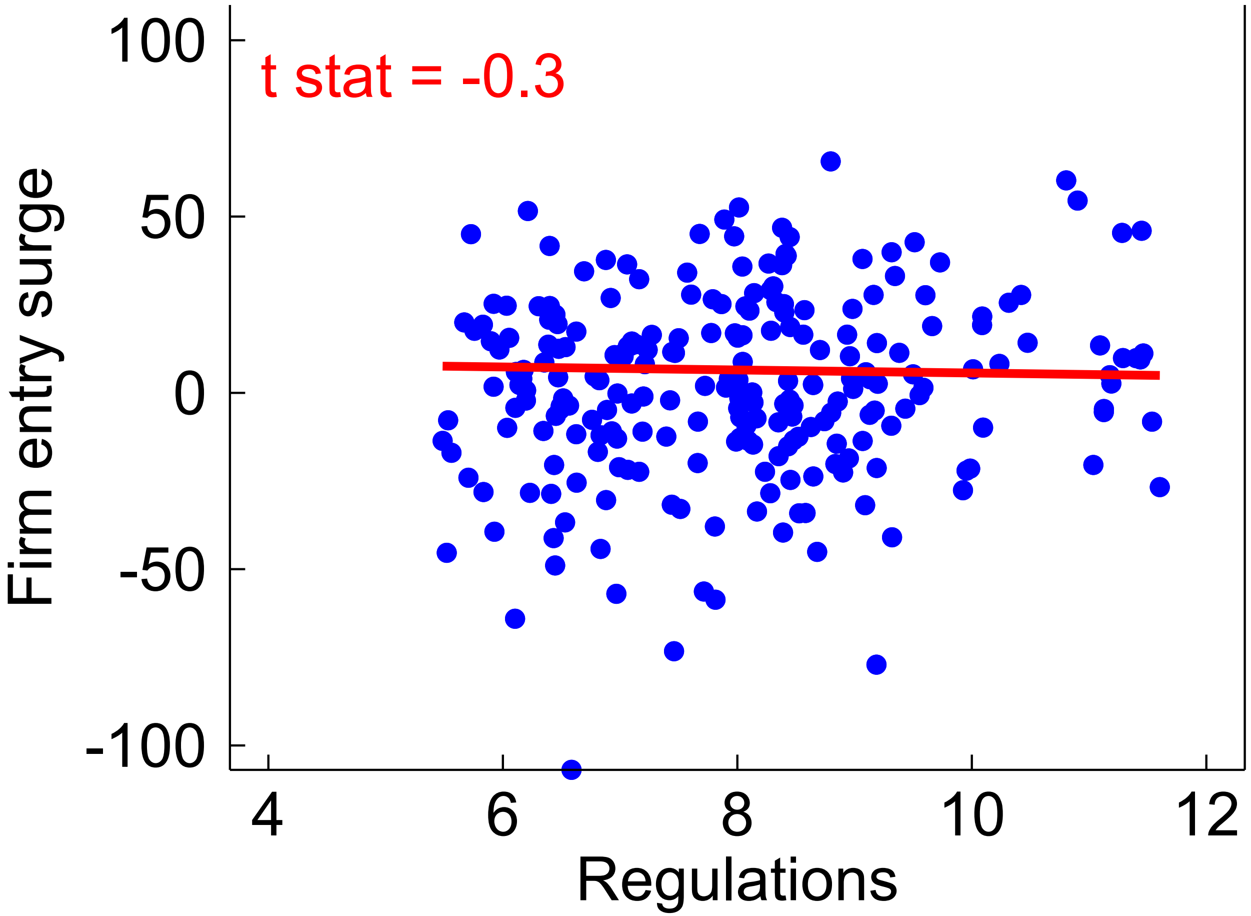
- **Result:**
  - No relationship in weighted regressions → no aggregate story
  - Not shown: Positive relationship in unweighted regressions → more regulated industries saw *larger* entry surge



Note: Entry surge in logs, 2021-22 vs. 2013-2019.
Regression line weighted by firm count.
Source: BDS, RegData, author calculations.

4. Labor market dynamics: “The Great Resignation”

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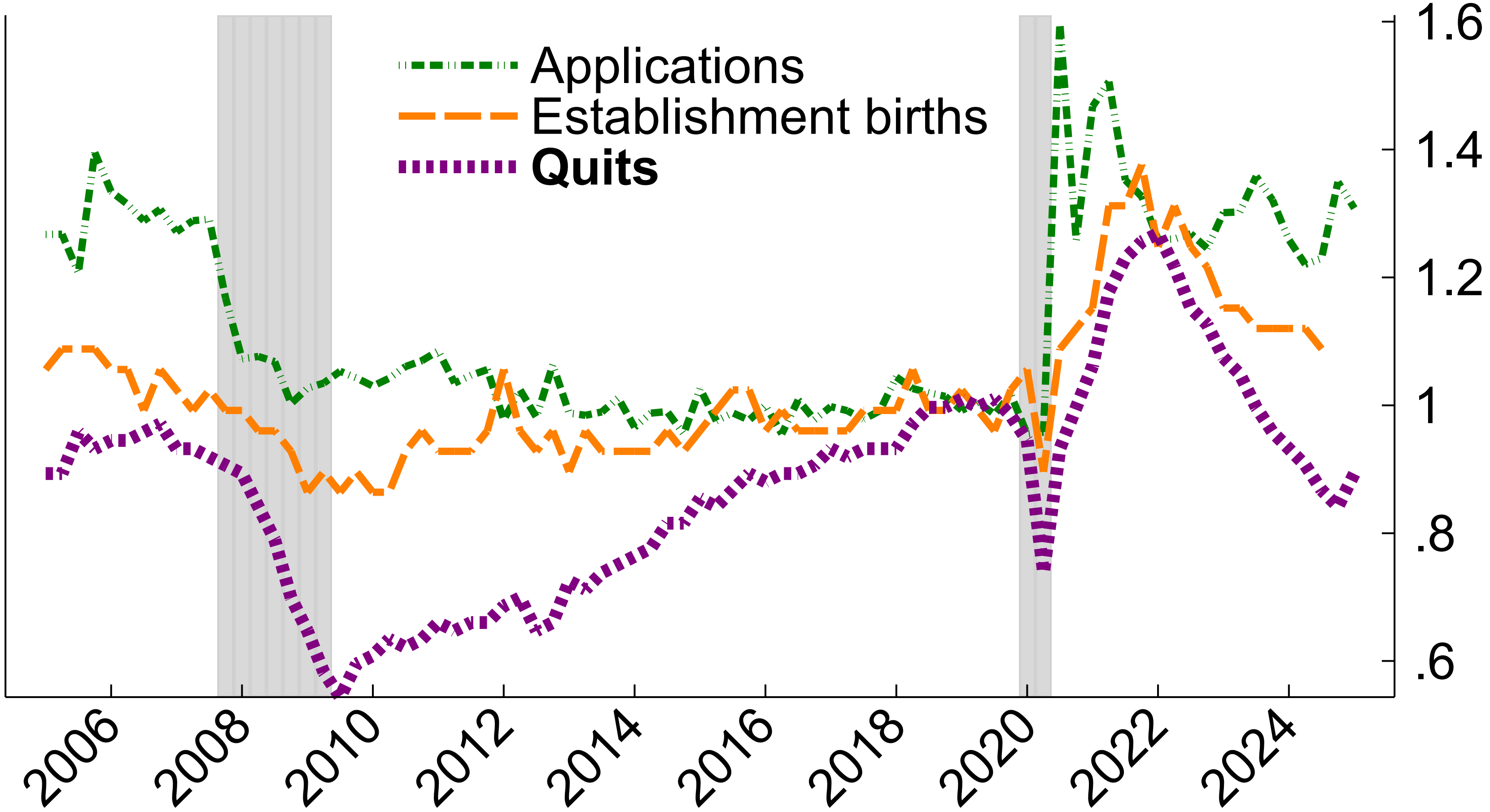
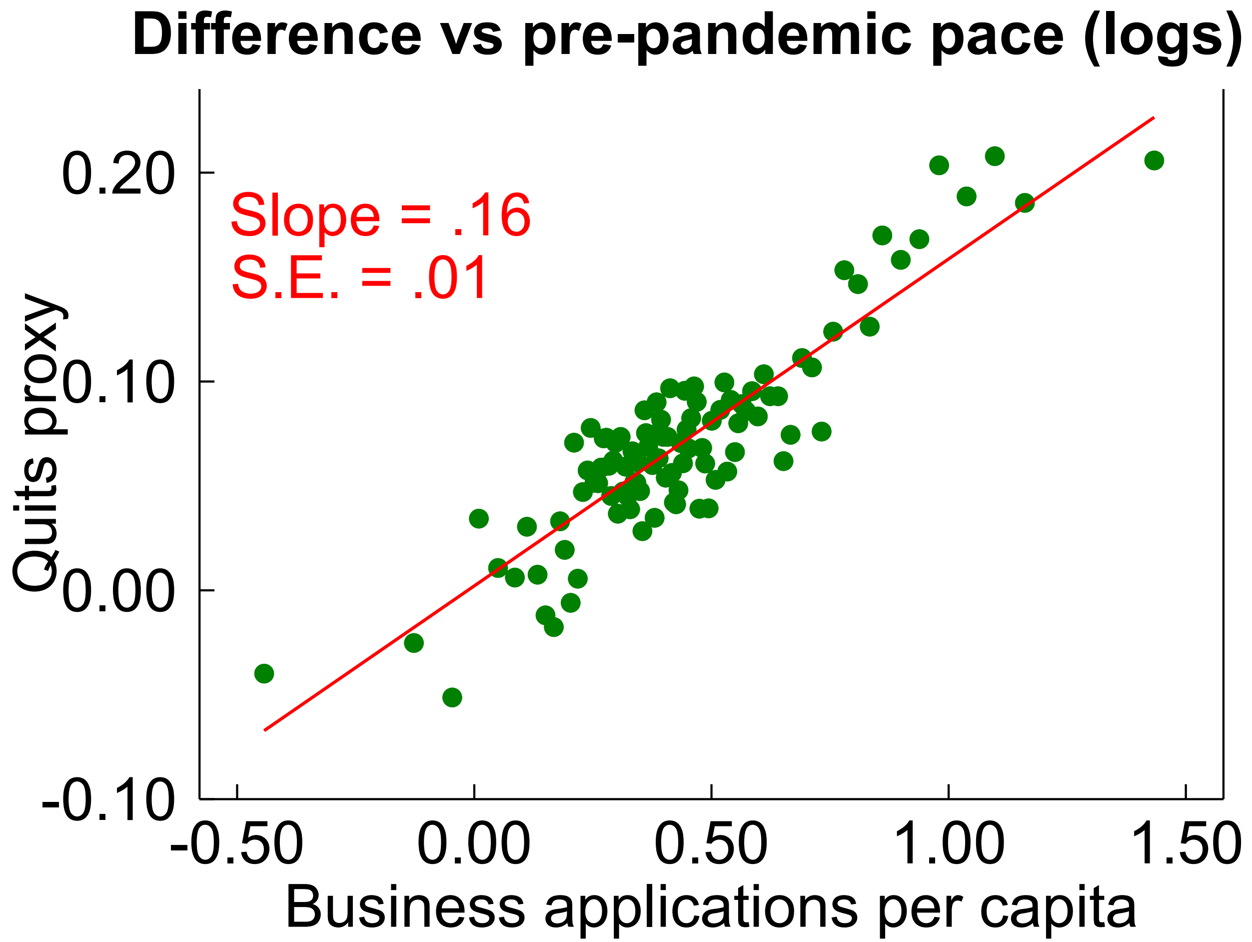
Business entry and worker quits (2019 = 1)



- The Great Resignation
- **Quits** surge and recede with **establishment births**
- Are the two related?

4. Labor market dynamics: Quitting to opportunity?



Note: 2020-2023 vs 2010-2019. County-level binscatter. Quits proxied by QWI excess separations.

- Counties with big quits surge are the counties with big business applications surge
- Not shown: Correlation for "layoffs" much weaker.
- What is the story?
  - Likely: Many workers quitting to join (or start) new businesses
  - Not/less likely: Business formation surge explained by layoffs and weak labor market

What does America's pandemic entry surge *mean?*

- Never count out (potential) entrepreneurs!
- Industry and geography stories:
  - Pandemic entry surge was part of the economy's adjustment to changing patterns of consumption, work, and life
  - Tech entrepreneurship (and firm expansion) likely related to remote work, AI developments
- Policy: Macroeconomic policy likely helped
- Labor market story: The Great Resignation may have partly been workers flowing to new firms (as early employees or founders)

Open questions

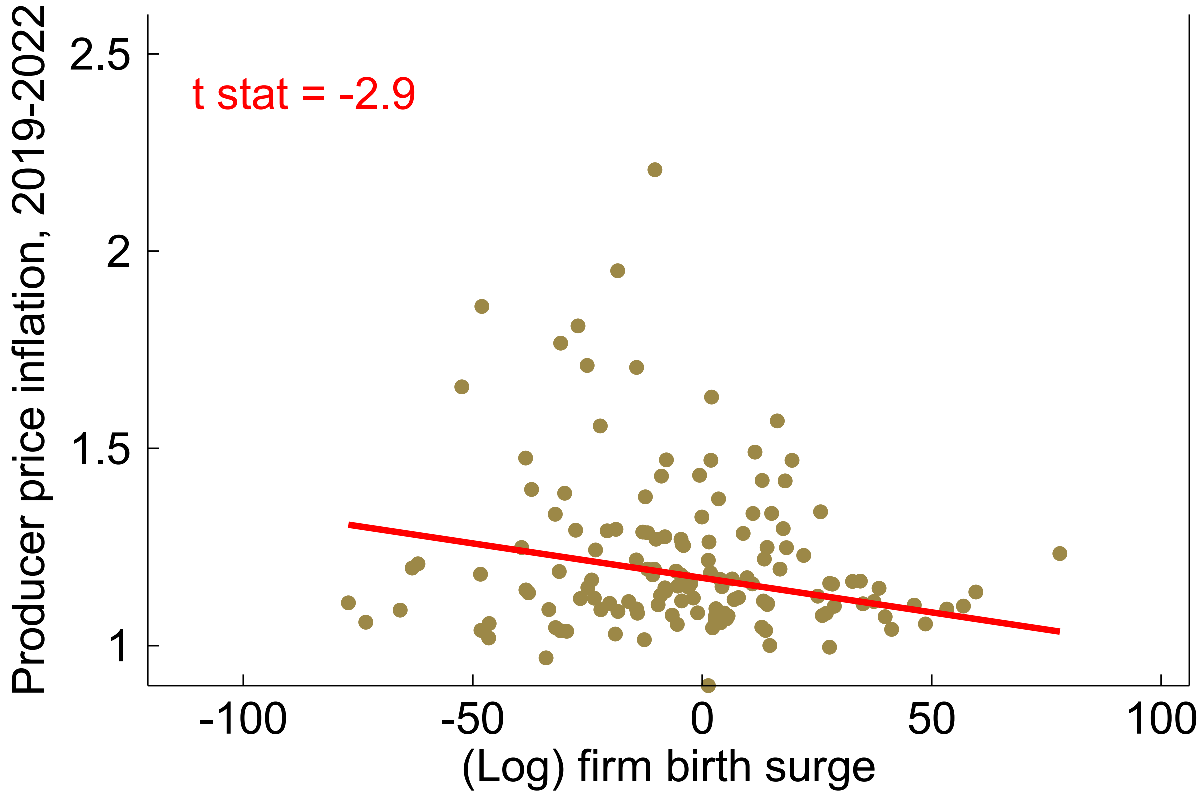
- What explains the **cross-country variation**? Some countries surged (U.S., Australia, France); most didn't.
  - U.S. policy evidence, AI may be clues
- Need more **policy analysis**! Country, state variation?
- What was entrepreneurs' role in post-pandemic **supply chain** crisis, **inflation**?
  - Entry surge in transportation, freight → entrants may have helped
  - Preliminary work: **Inflation** looks lower in high-entry industries
  - To do: Zero in on supply chain industries
- What about "**gig workers**" and nonemployer businesses? In progress…
- How has the surge **changed over time**?
  - Recent seeming divergence between applications and employer entry
  - Macro policy tightening
  - Recent **trade shocks**
- Eat your vegetables: Are our **statistical resources** up to the task?

Thanks

Extra slides

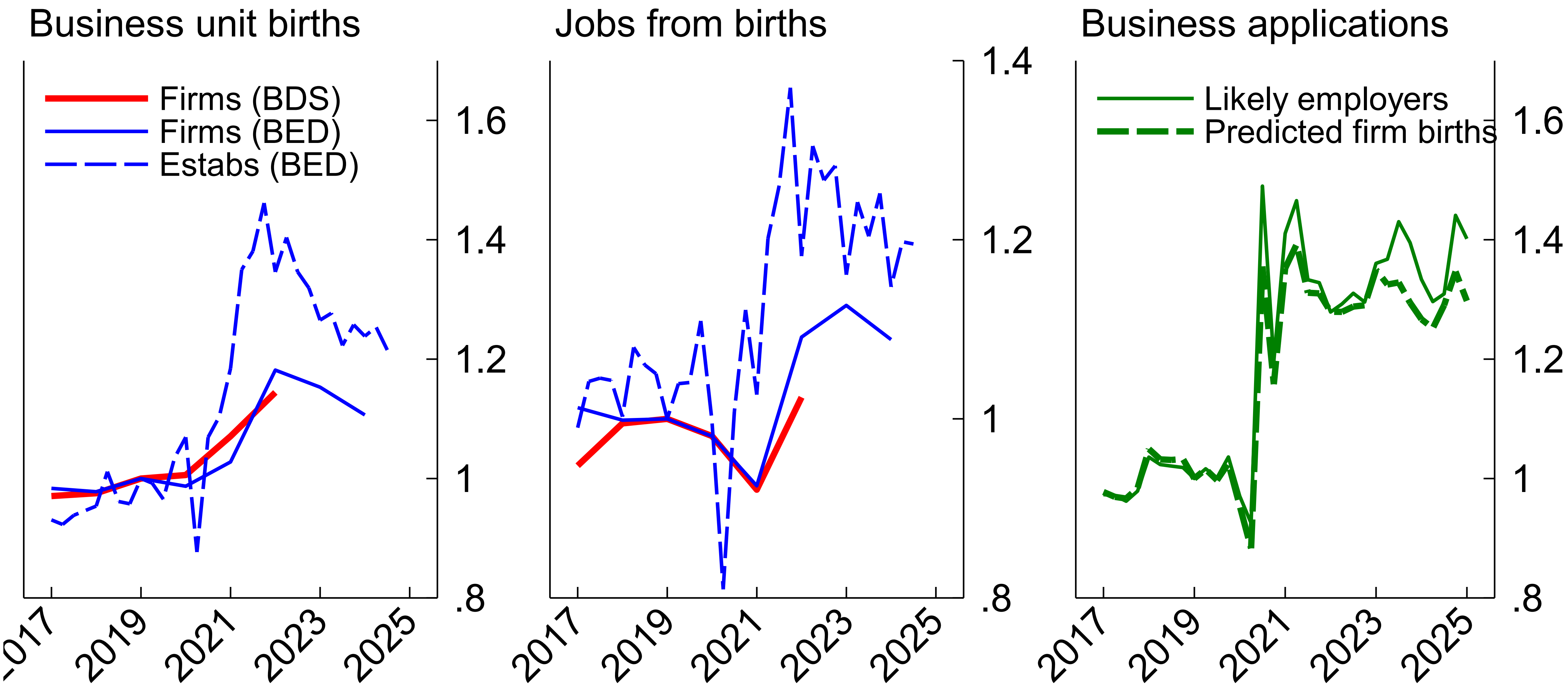
Inflation (preliminary and incomplete)*

- Could imagine causality either way:
  - Industries with high demand, inflation → entry incentive
  - Entry boosts industry supply → lower inflation
- Compare industry-level producer price inflation to firm entry, 2019-2022
- Regression weighted by employment (better weights in progress)



*Joint work with Jacob Williams (FRB)

Surge continues but may be cooling

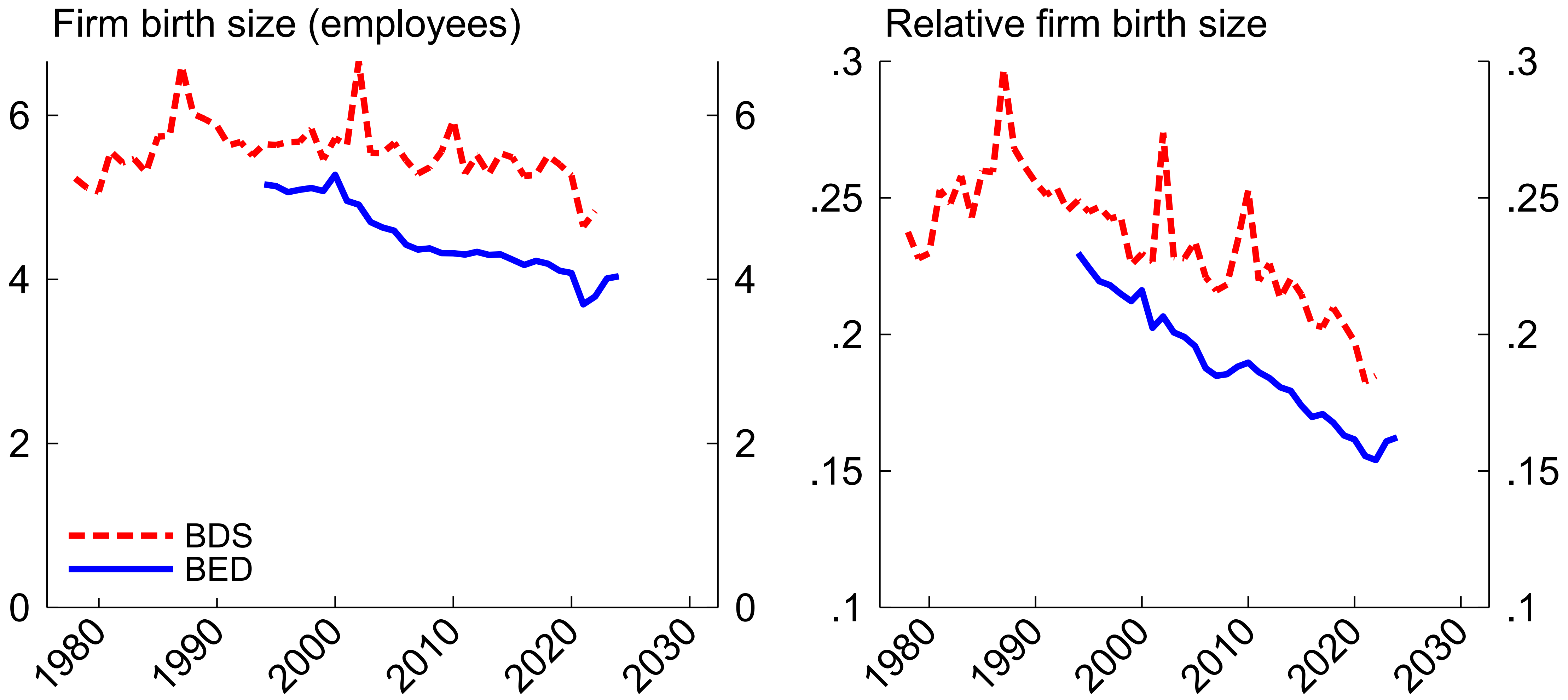


Note: 2019:Q1 = 1. Predicted firm births within 8 quarters.
Source: Census Bureau BDS, BLS BED, Census Bureau BFS.

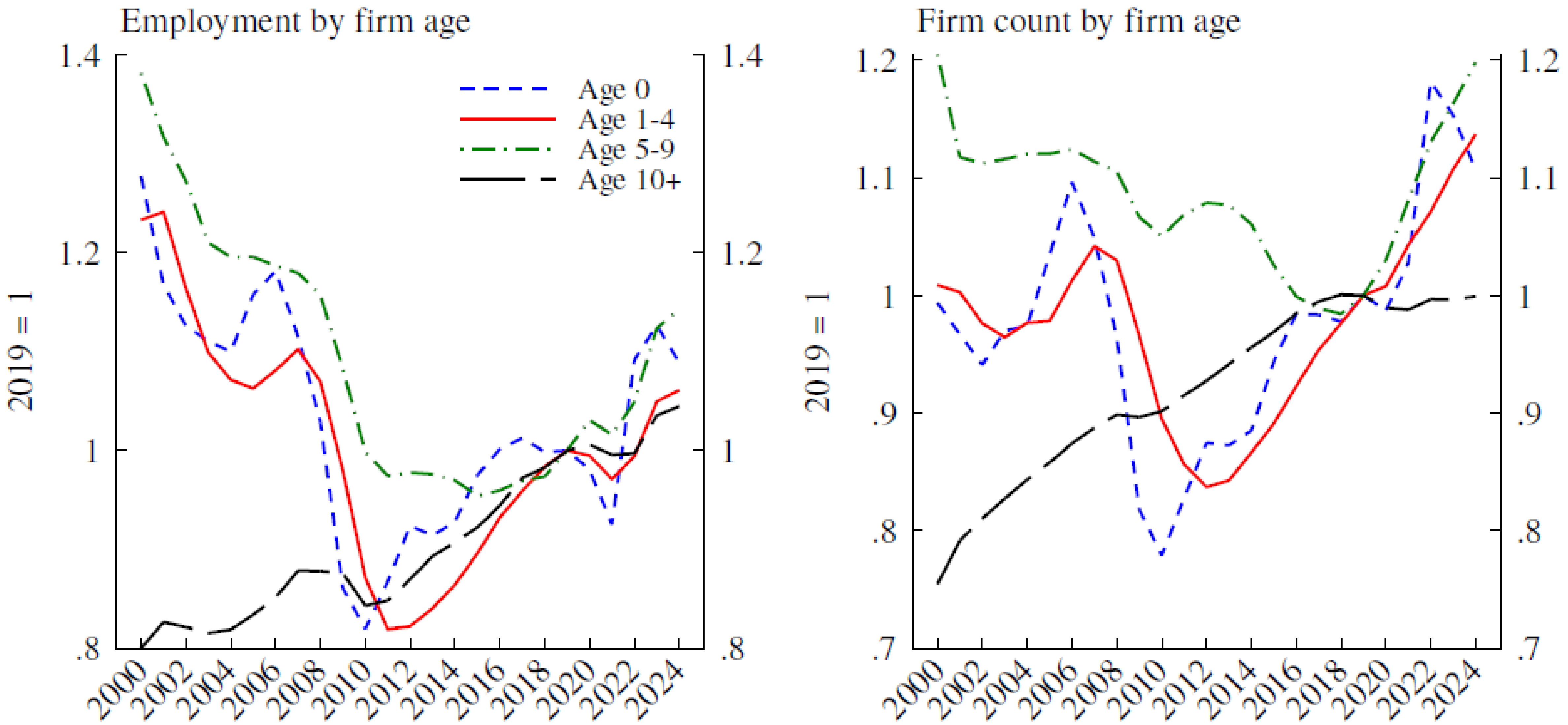
Firm exit

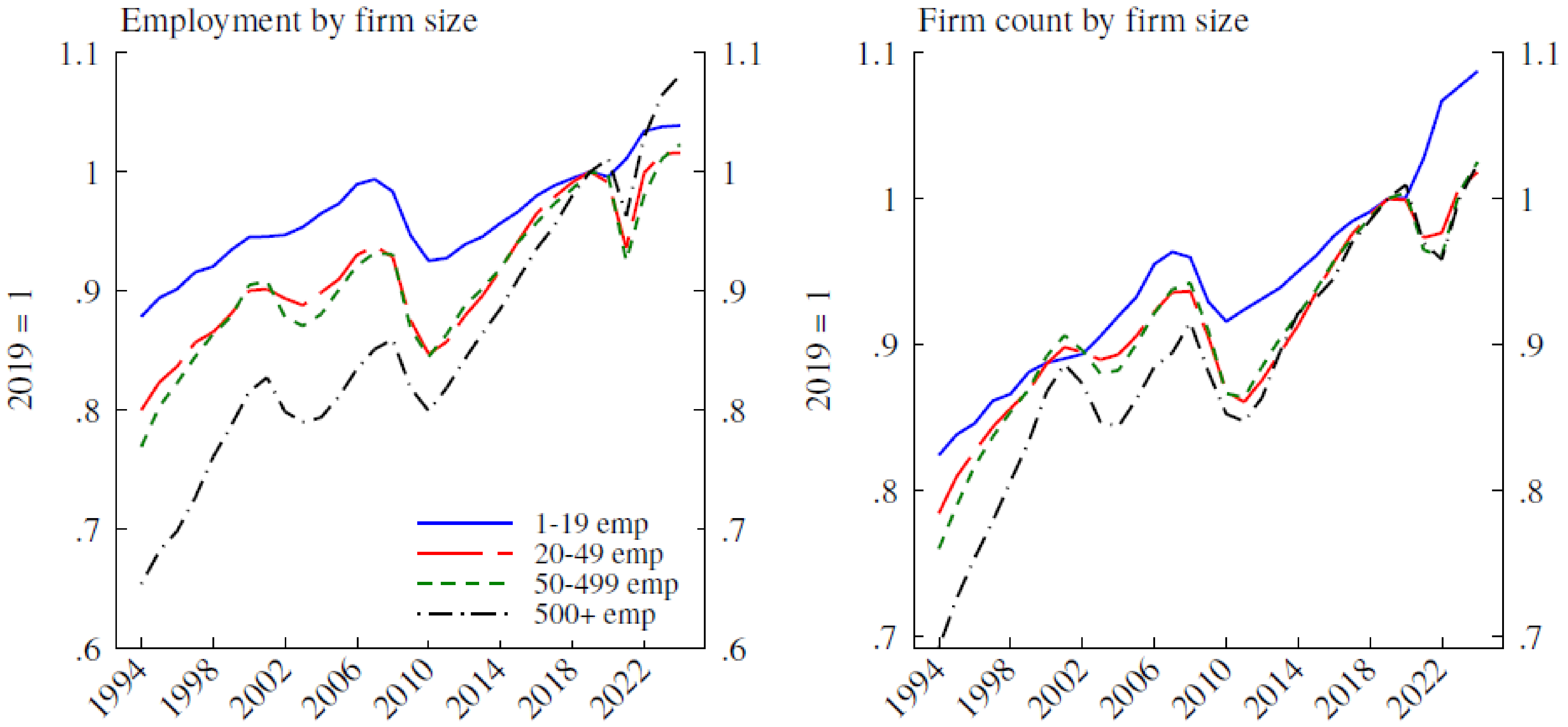
Note: Firm entry rates. Right panel uses DHS denominator.
Source: Business Dynamics Statistics (BDS) and Business Employment Dynamics (BED).

Entrant size



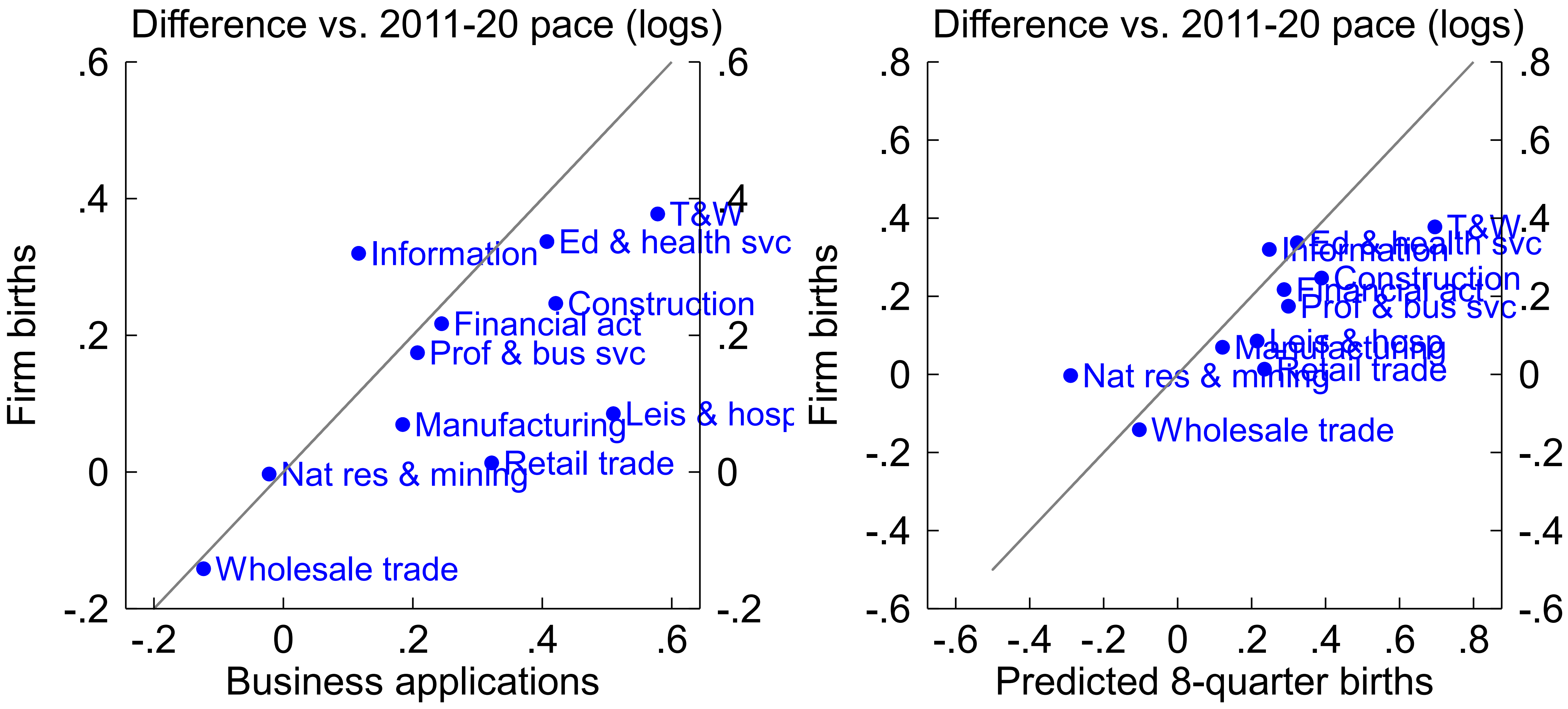
Note: Average size in first year (left); relative to incumbent average size (right).
Source: Business Dynamics Statistics (BDS) and Business Employment Dynamics (BED).


Employment by firm age
Age 0
Age 1-4
Age 5-9
Age 10+
2019 = 1
1.4
1.2
1
.8
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Firm count by firm age
2019 = 1
1.2
1.1
1
.9
.8
.7



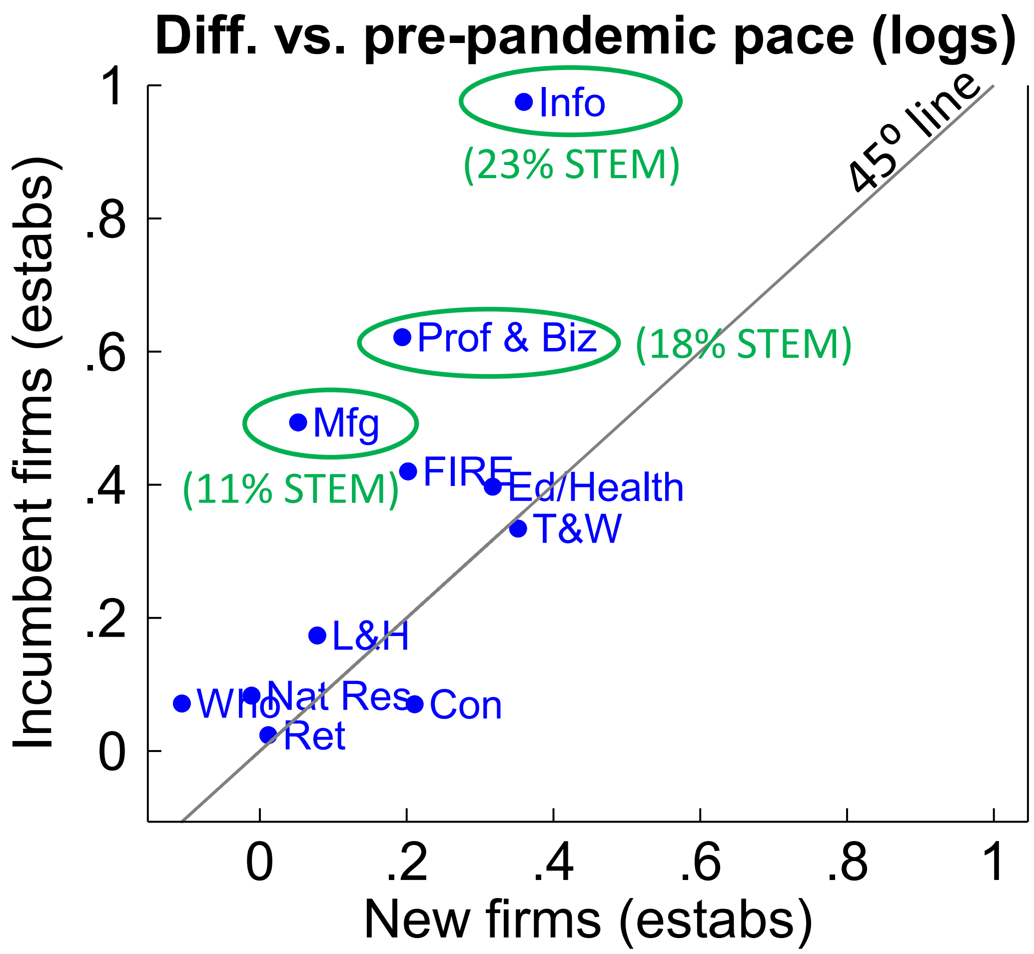
Note: March snapshots. For age classes above 0, employment measured as implied quarterly DHS denominator.
Source: Business Employment Dynamics (BED).

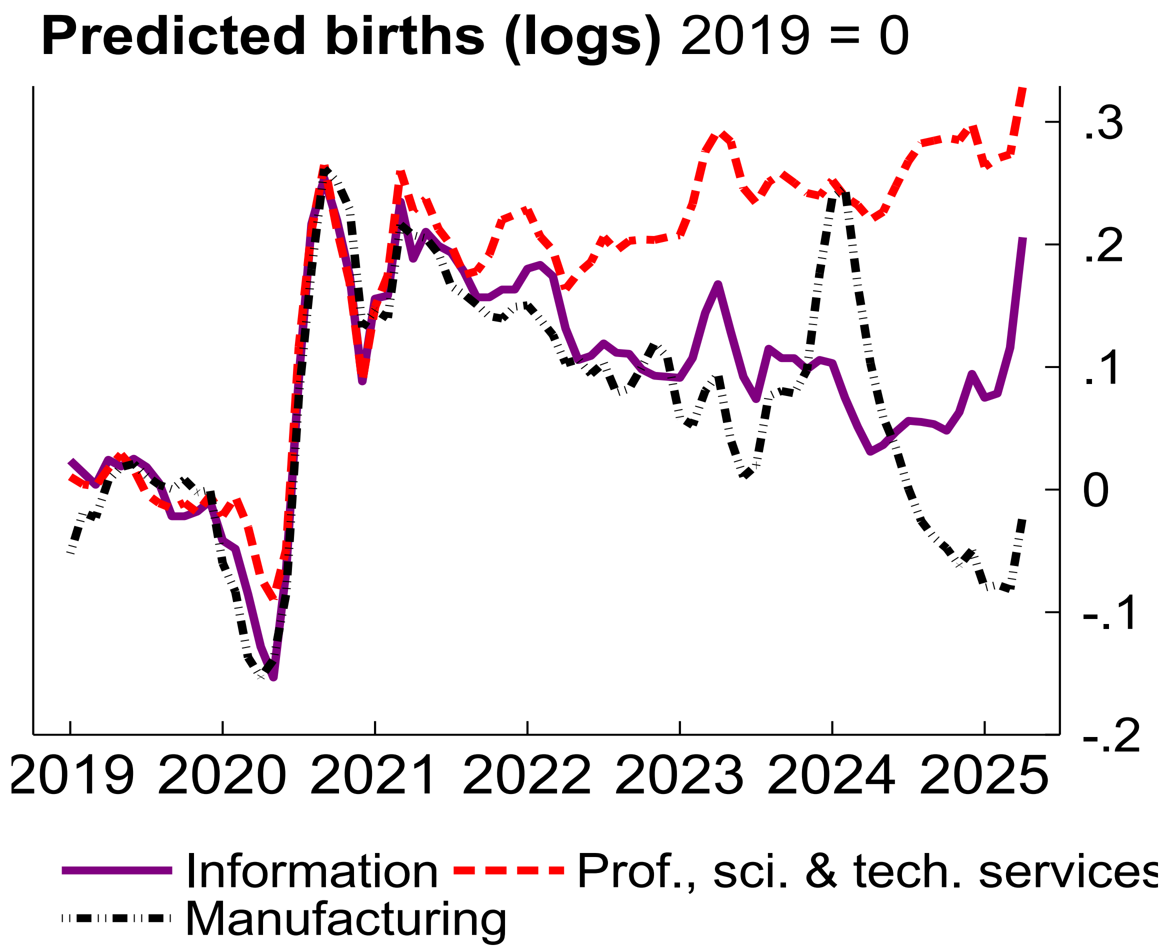
Transitions: application to firm birth



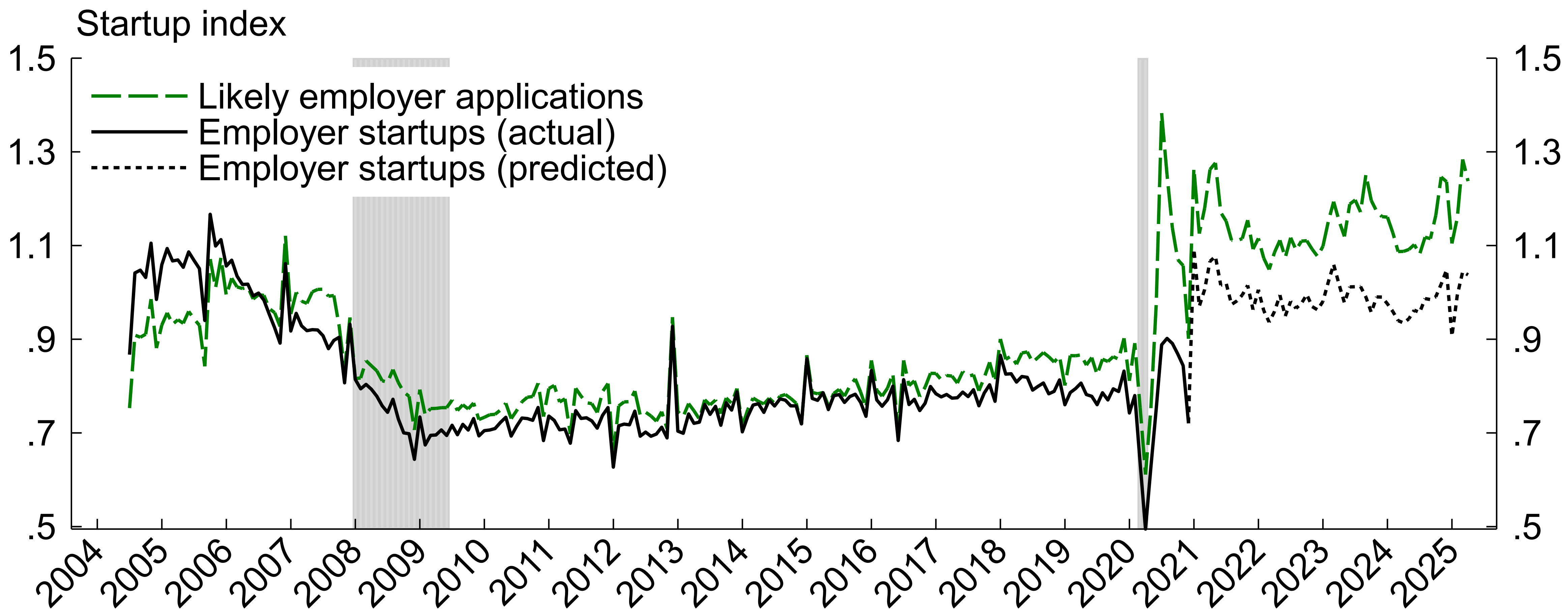
Note: 2021-2024. Solid line is 45-degree line. T&W is transportation & warehousing. Years end in March.
Source: Business Employment Dynamics (BED), Business Formation Statistics (BFS).

Tech estab. entry, incumbents vs. new firms

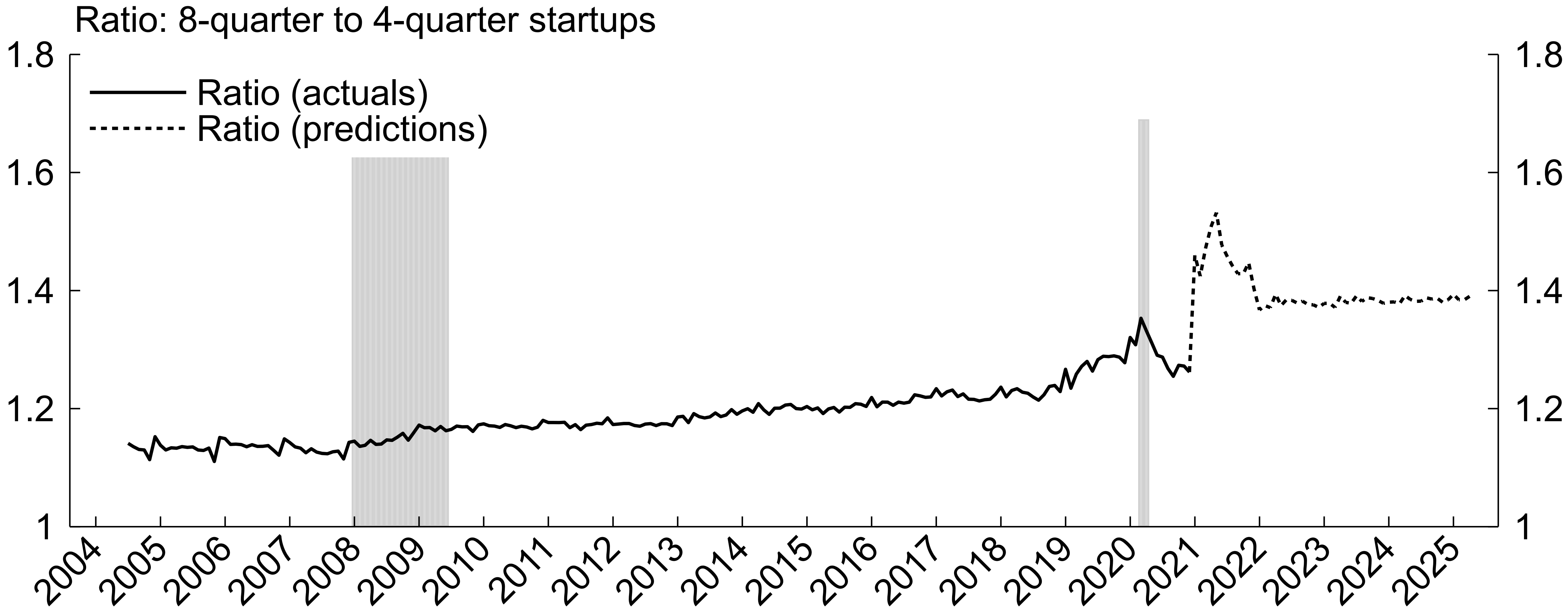




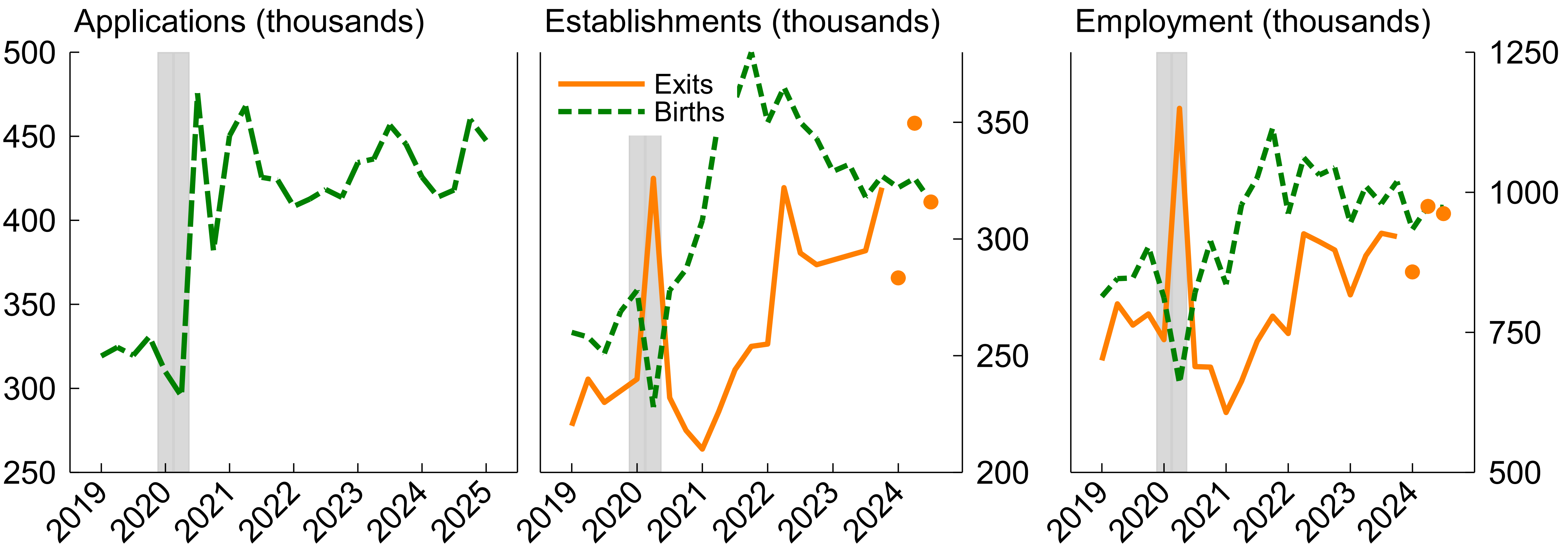
- **Top tech sectors** see more incumbent than new firm estab birth surge, but new firm surge apparent as well
- BFS predicted firm births in prof/sci/tech still elevated -> points to more tech firm births in future



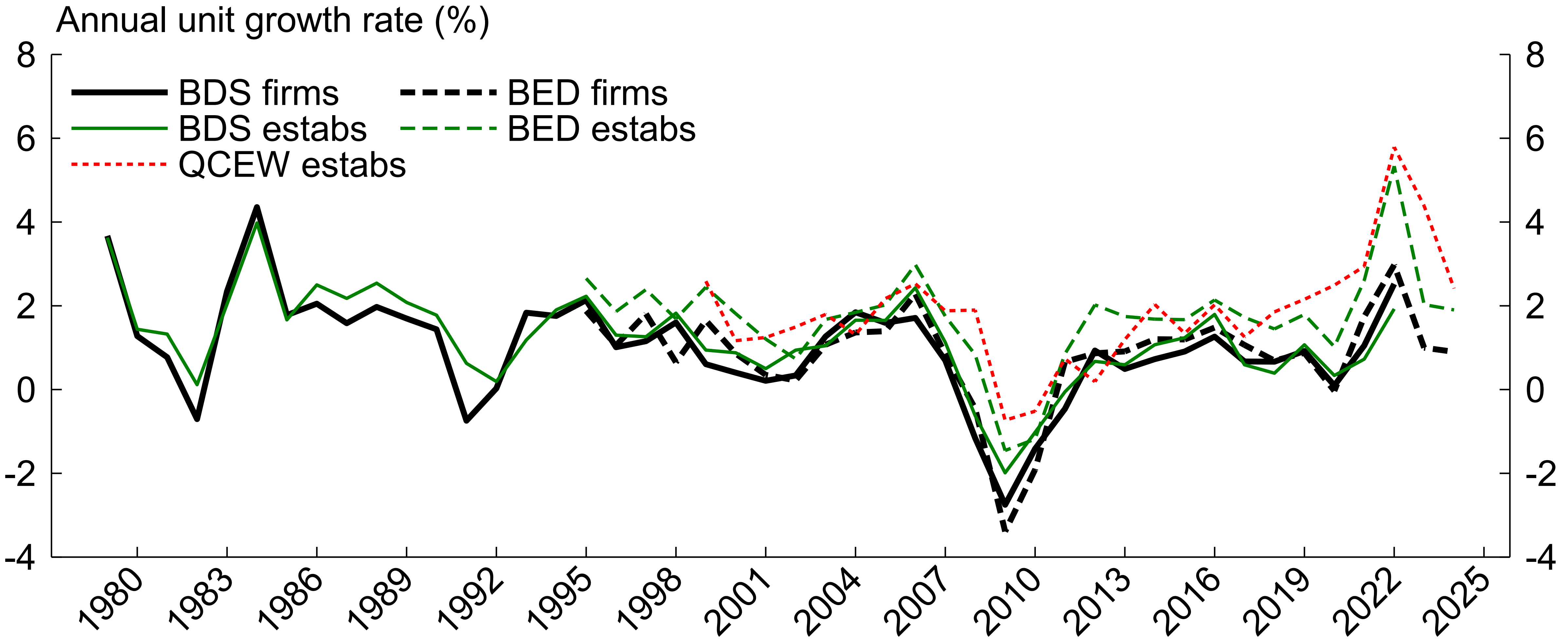
Note: Startups within 8 quarters. Seasonally adjusted. Normalized by average 2006 levels.
Shaded areas indicate NBER recession dates.
Source: Census Bureau Business Formation Statistics.



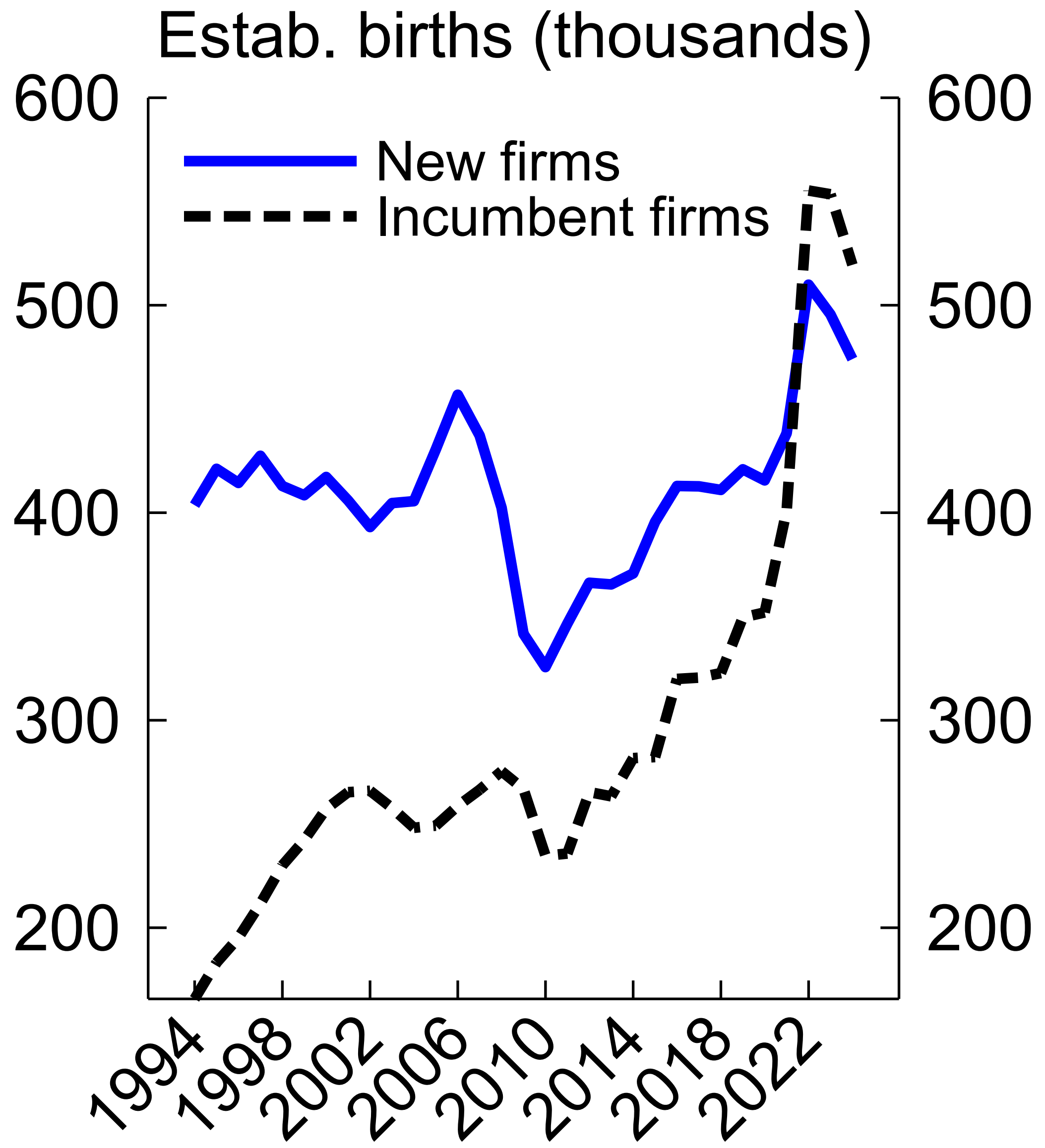
Note: Ratio of startups within 8 quarters of application to startups within 4 quarters of application. Seasonally adjusted before calculation. Shaded areas indicate NBER recession dates.
Source: Census Bureau Business Formation Statistics.

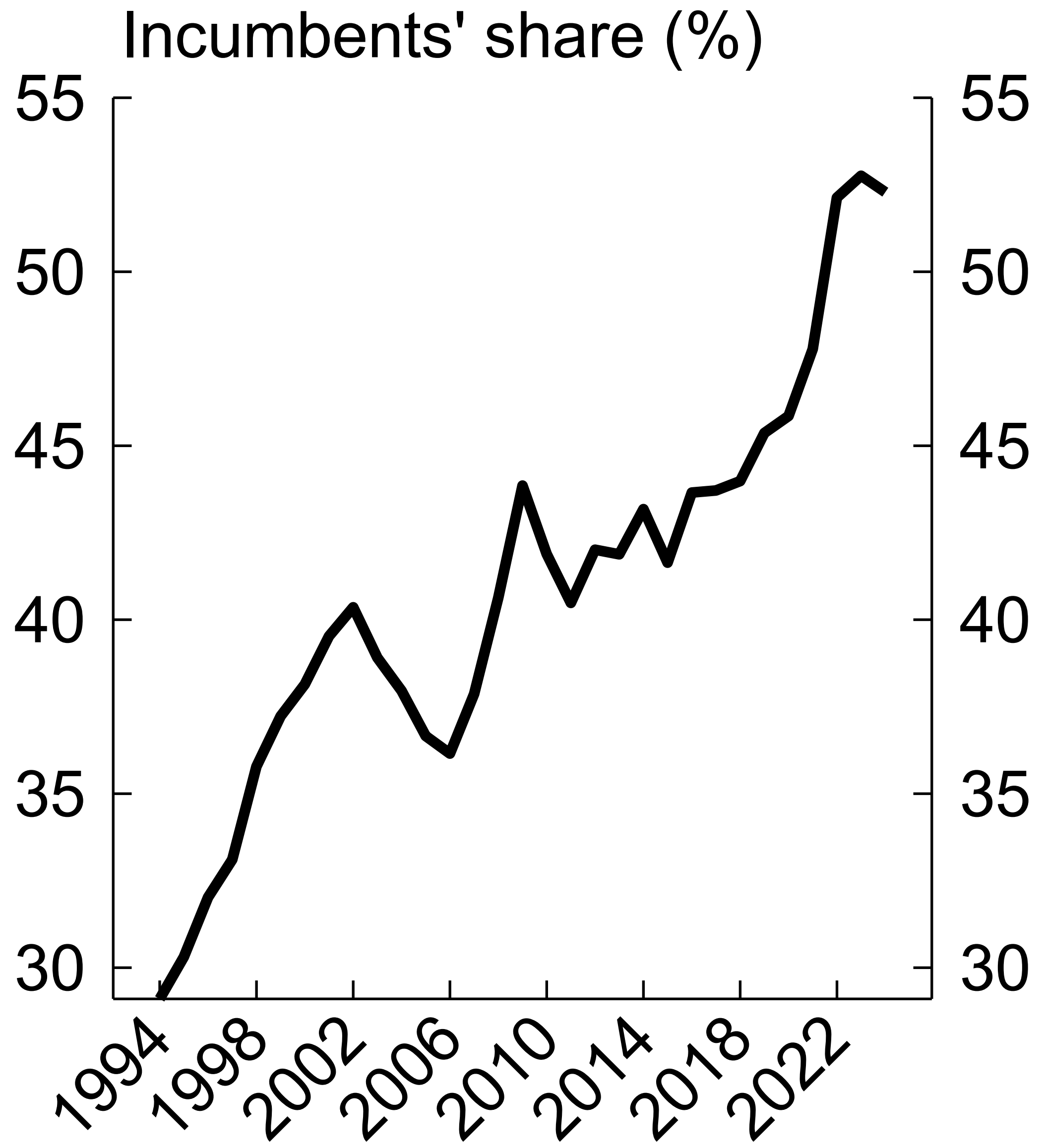


Note: Seasonally adjusted. Y axes may not start at zero. Shaded areas indicate NBER recession dates. High-propensity applications. Exits after 2023q4 projected based on most recent share of exits in closures (orange dots).
Source: Census Bureau Business Formation Statistics (BFS) and BLS Business Employment Dynamics.



Note: Annual DHS growth rate of unit counts, Q1 versus year earlier.
Source: BDS, BED, QCEW.





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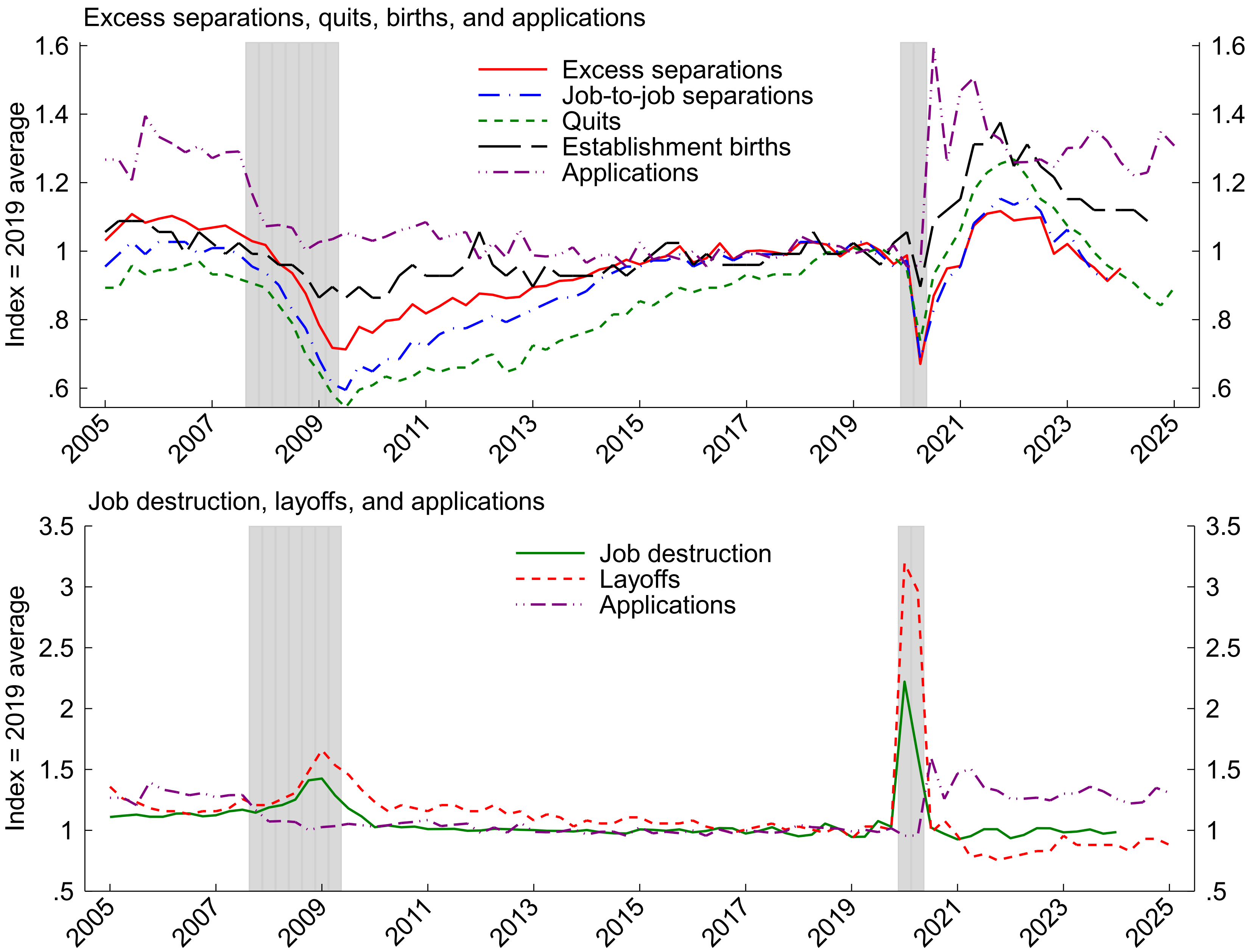
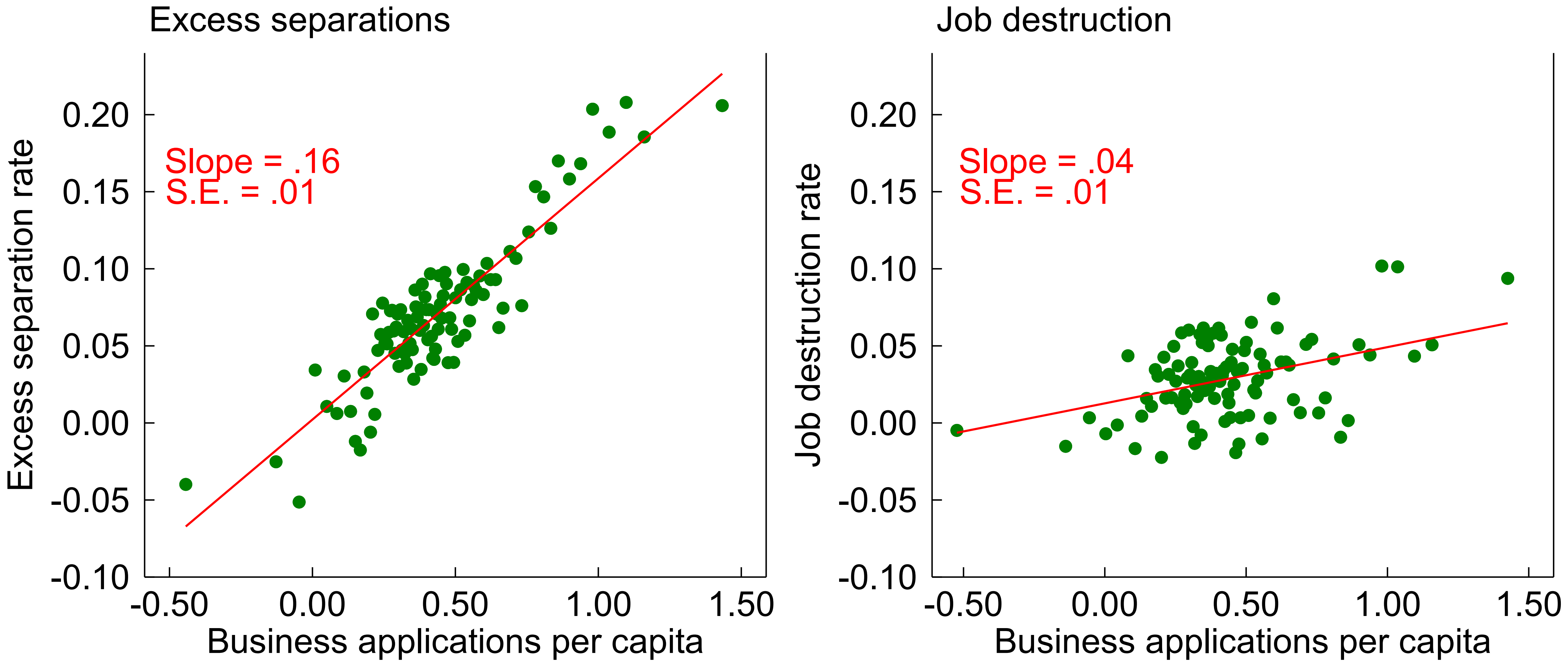
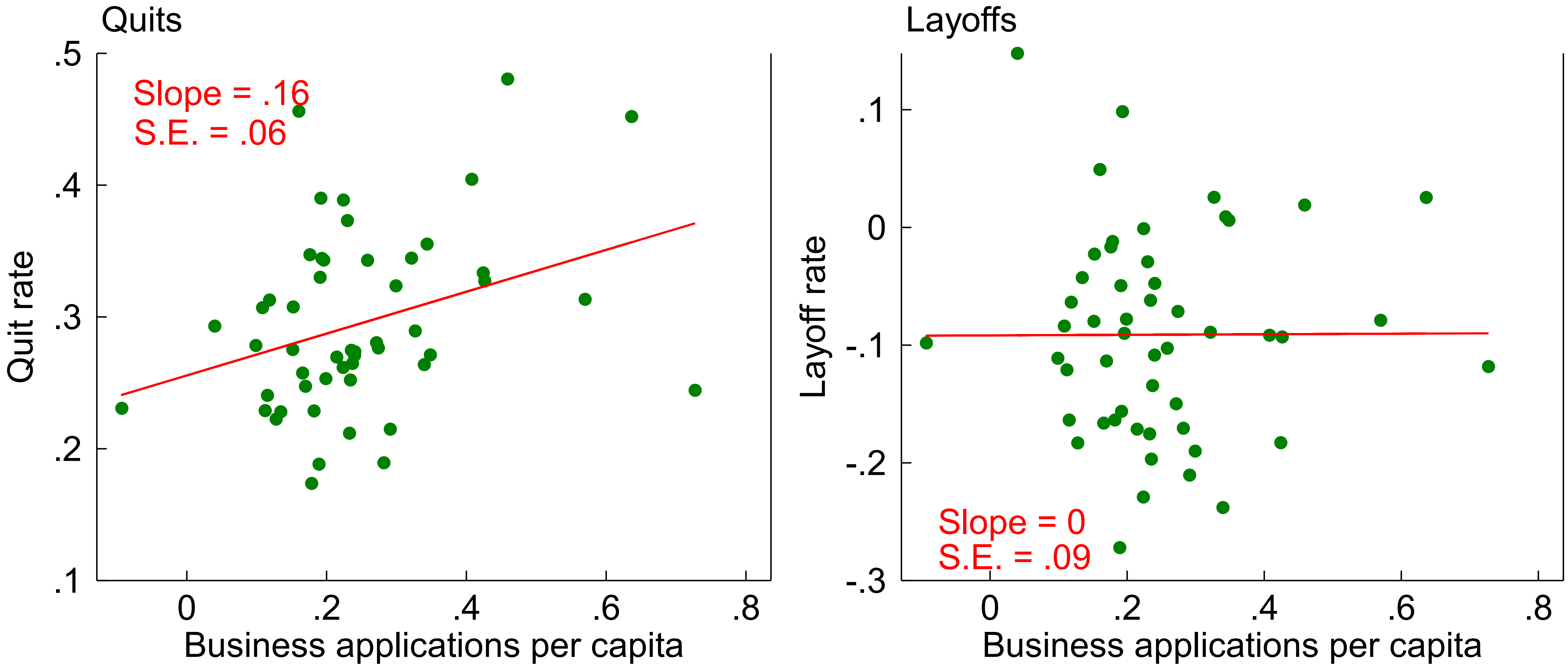
Source: Business Employment Dynamics (BED).



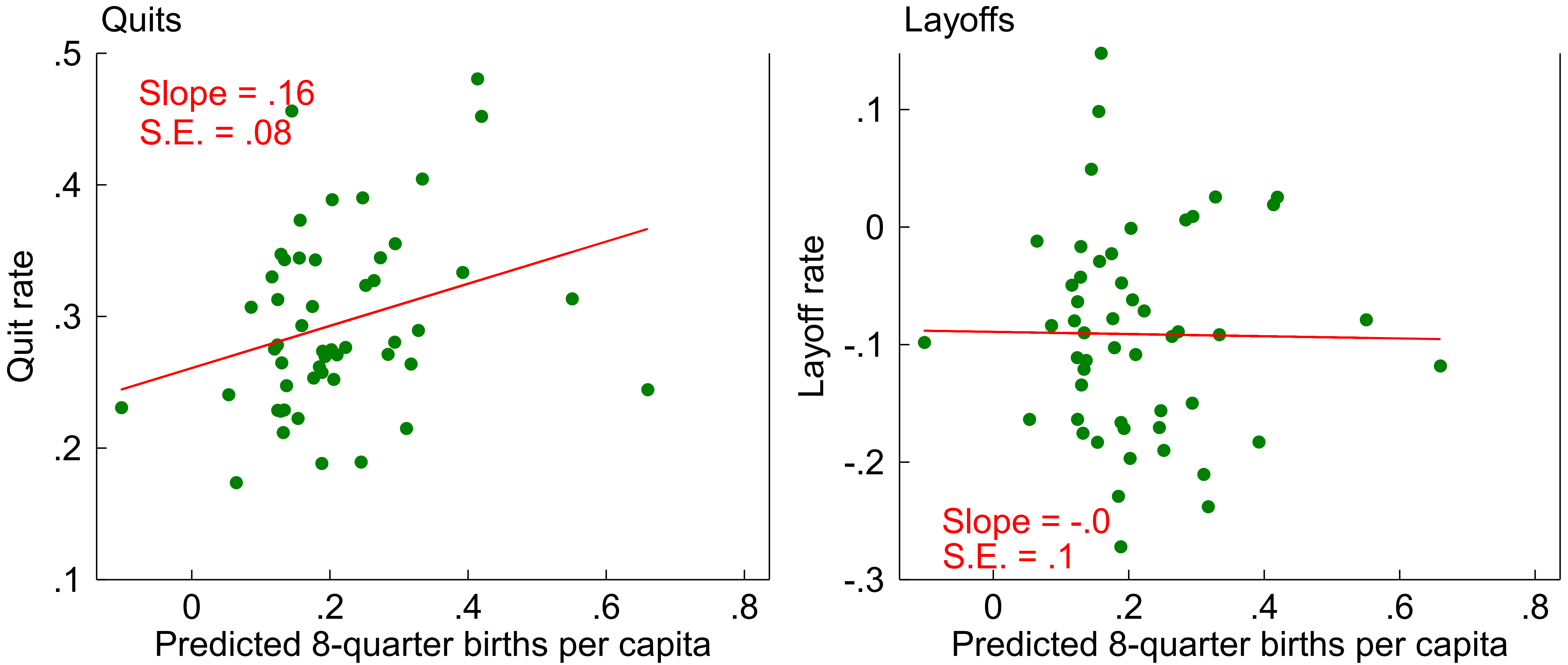
Note: Index of series expressed relative to employment or, for births, to establishments; seasonally adjusted.
Applications are likely employers (HBA). Shaded areas indicate NBER recession dates.
Source: QWI, JOLTS, BED, BFS, and CES.



Note: County-level log differences of 2020-2023 vs. 2010-2019 seasonally adjusted pace. Red line is regression line with reported slope and standard error. Binscatter with 100 bins.
Source: Quarterly Workforce Indicators (QWI), Business Formation Statistics (BFS).



Note: State-level log differences of 2020-2023 vs. 2010-2019 seasonally adjusted pace. Red line is regression line with reported slope and standard error. Data through March 2025.
Source: JOLTS, Business Formation Statistics (BFS).



Note: State-level log differences of 2020-2023 vs. 2010-2019 seasonally adjusted pace. Red line is regression line with reported slope and standard error. Data through December 2023.
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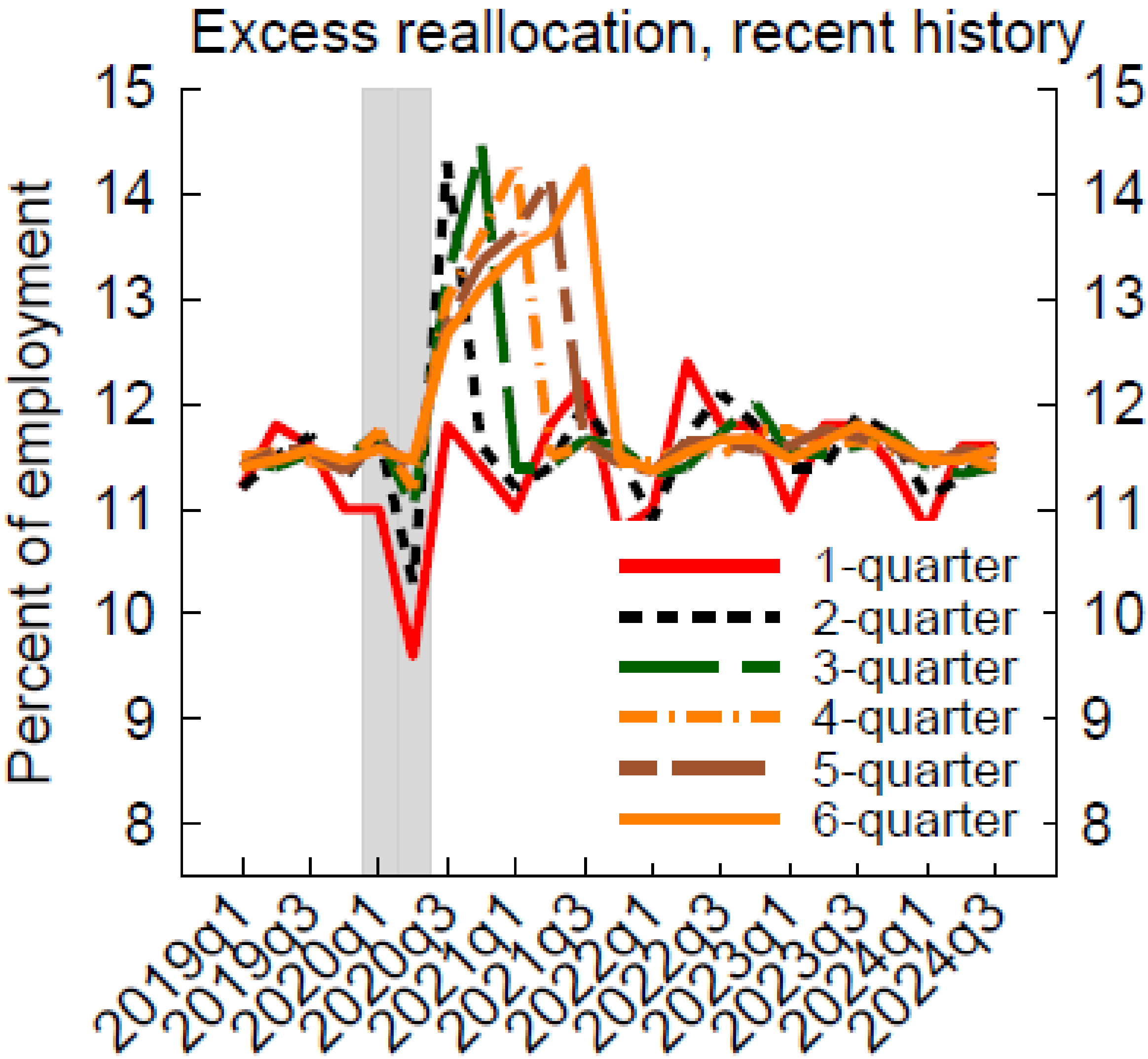
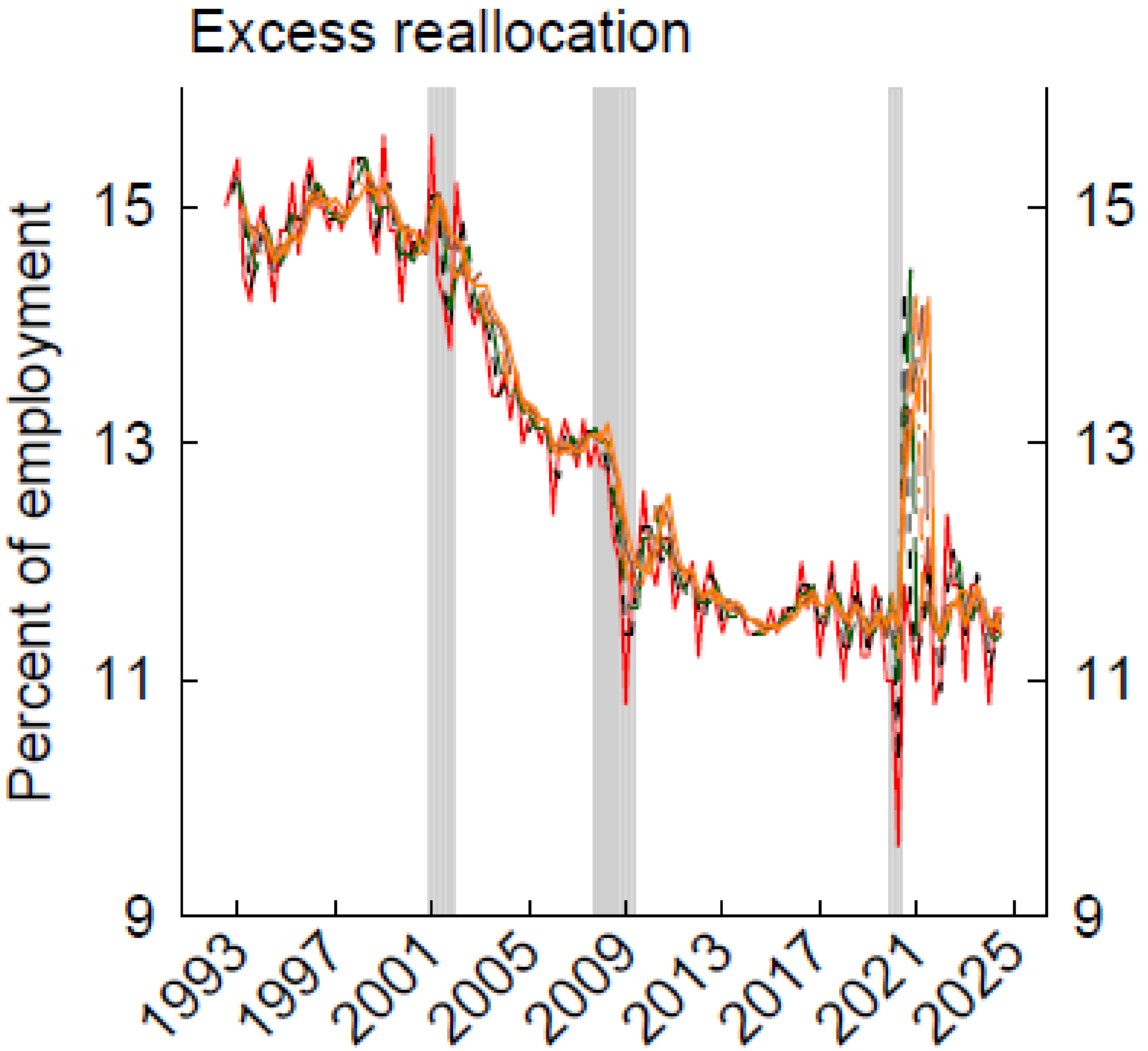


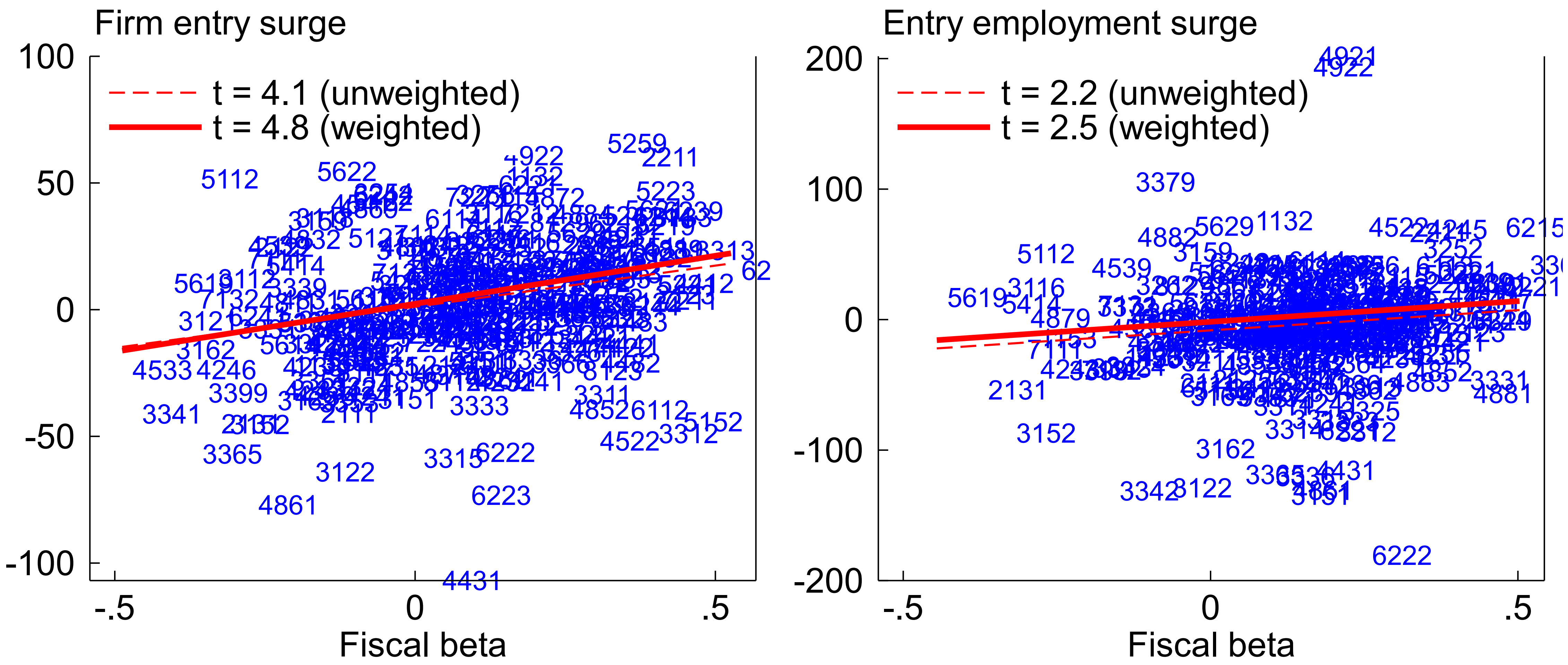


Note: Reallocation is JC+JD. Excess reallocation is JC+JD-|JC-JD|, with JC and JD averaged over indicated horizon. Seasonally adjusted. Shaded areas indicate NBER recession dates.
Source: Business Employment Dynamics (BED).

Joint regressions (t stats)

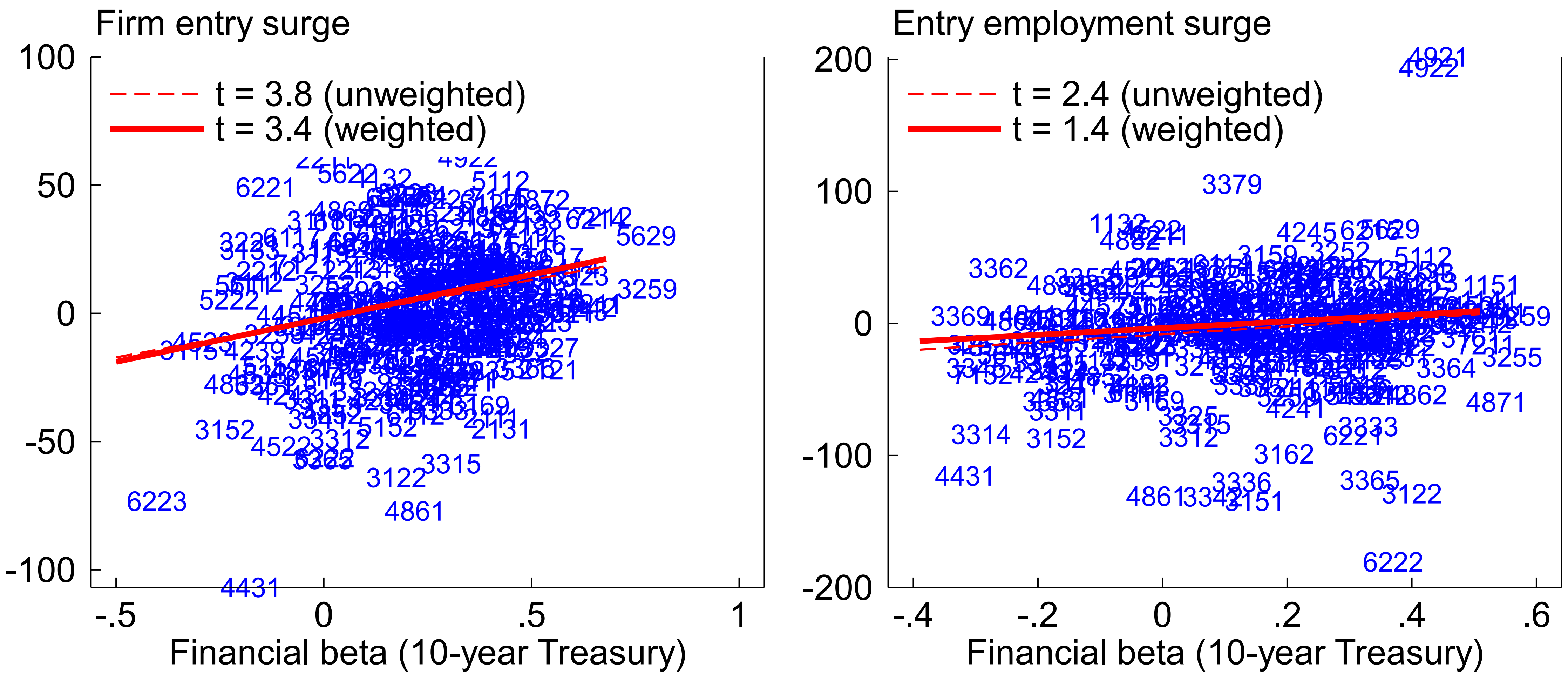
	Firm entry	Firm entry	Entry employment	Entry employment
Fiscal beta	**2.3**	**2.7**	1.8	-0.0
Financial: house price	1.7	**2.0**	1.9	0.9
Financial: Treasury	**2.2**	**2.2**	**3.0**	1.9
Regulation	**2.3**	-0.4	**2.2**	-0.2
R^2	.14	.21	.12	.25
Weighted?		Y		Y

Fiscal beta



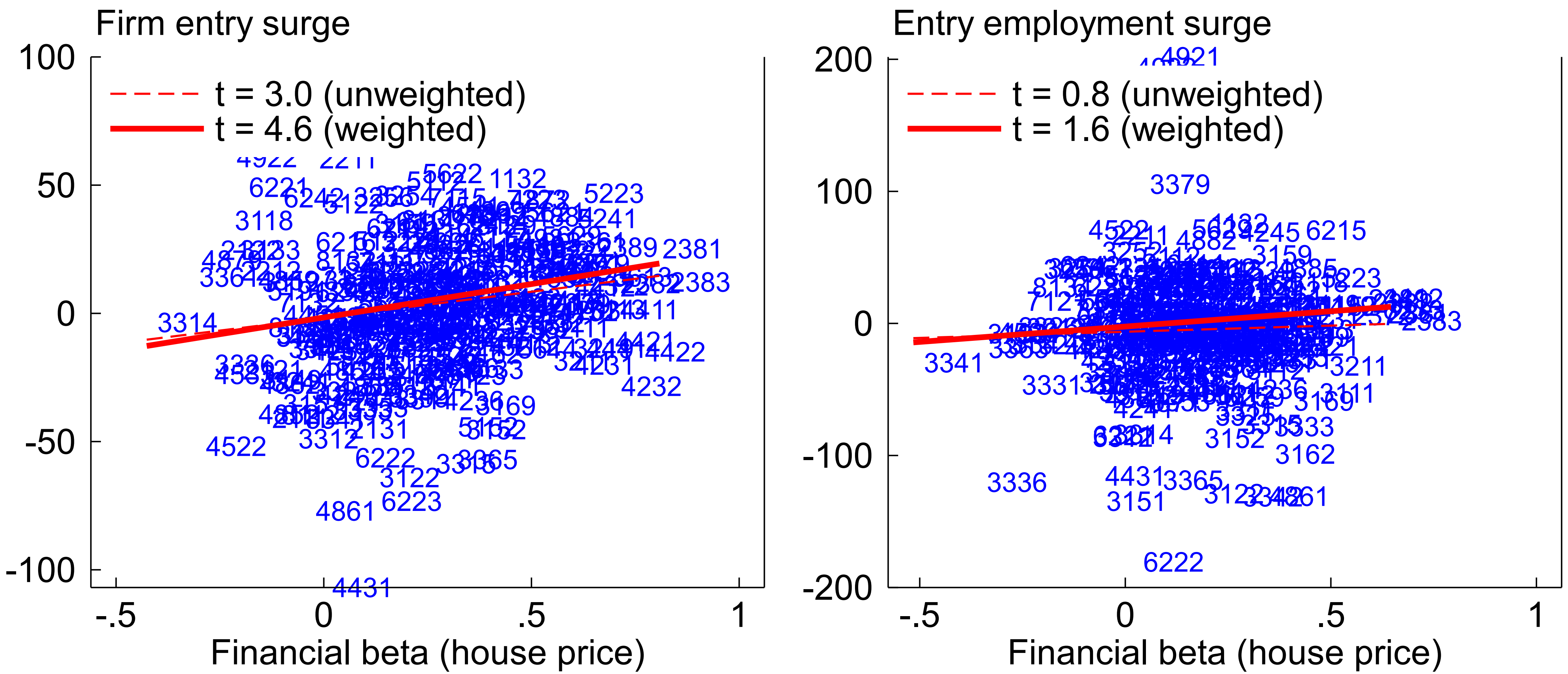
Note: Fiscal beta is correlation between industry entry and discretionary fiscal effect, 1978-2019.
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Cashin et al (2018), author calculations.

Financial beta: 10-year Treasury



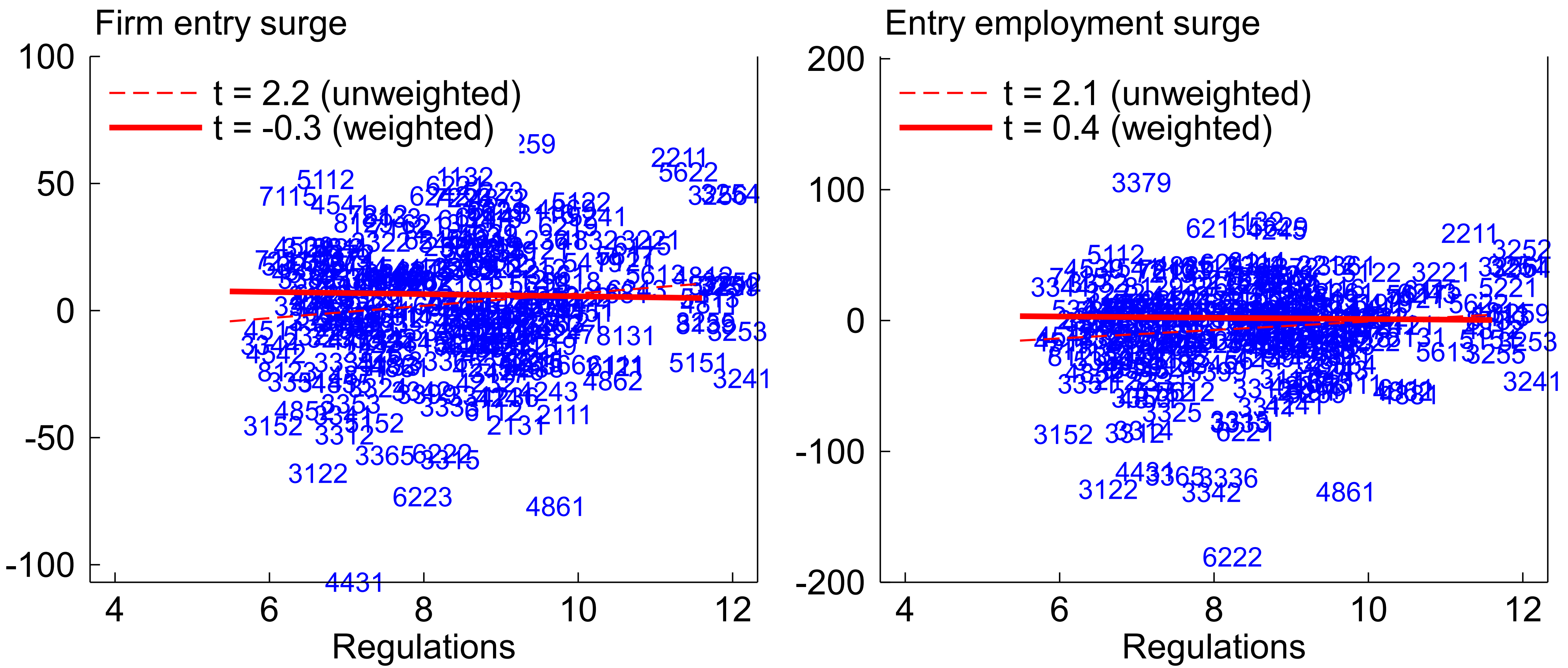
Note: Financial beta is correlation between industry entry and FCI-G 10-year Treasury component, 1991-2019.
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al (2023), author calculations.

Financial beta: House prices



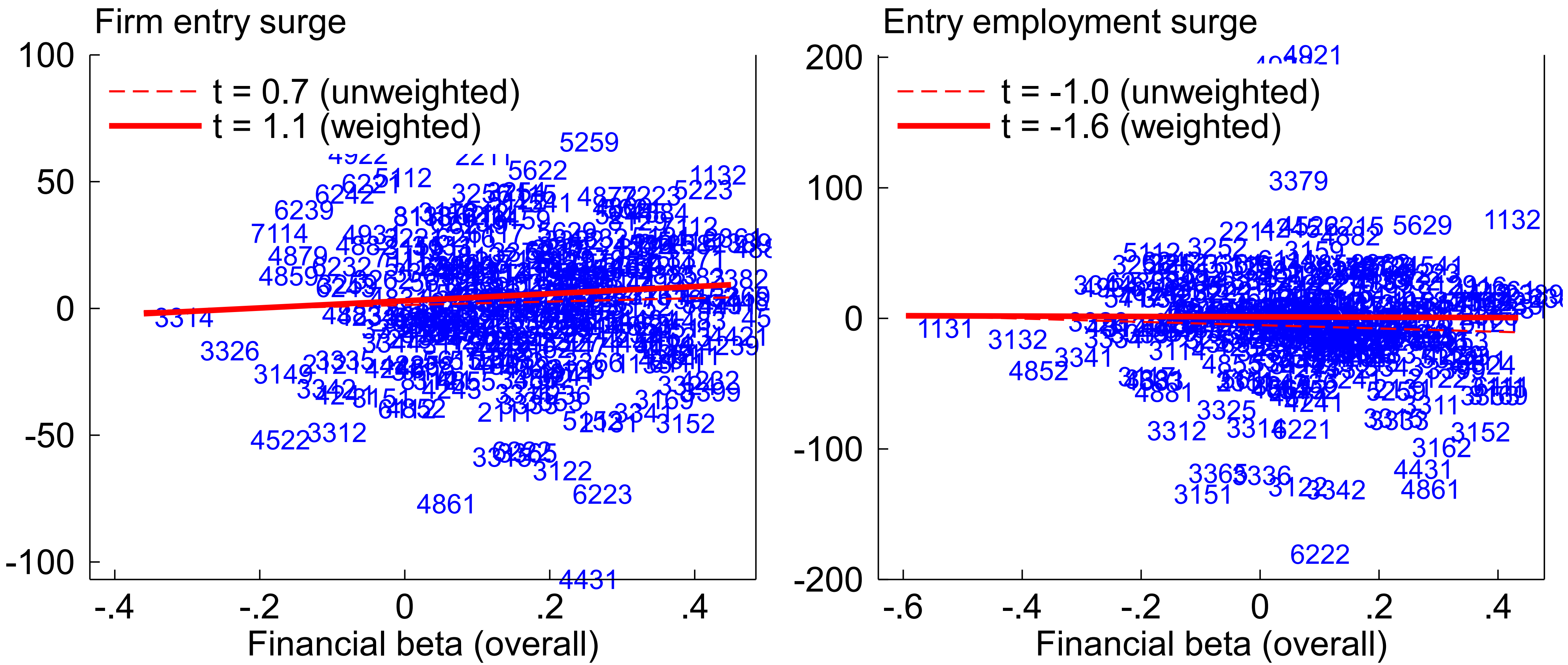
Note: Financial beta is correlation between industry entry and FCI-G house price component, 1991-2019.
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al (2023), author calculations.

Regulation



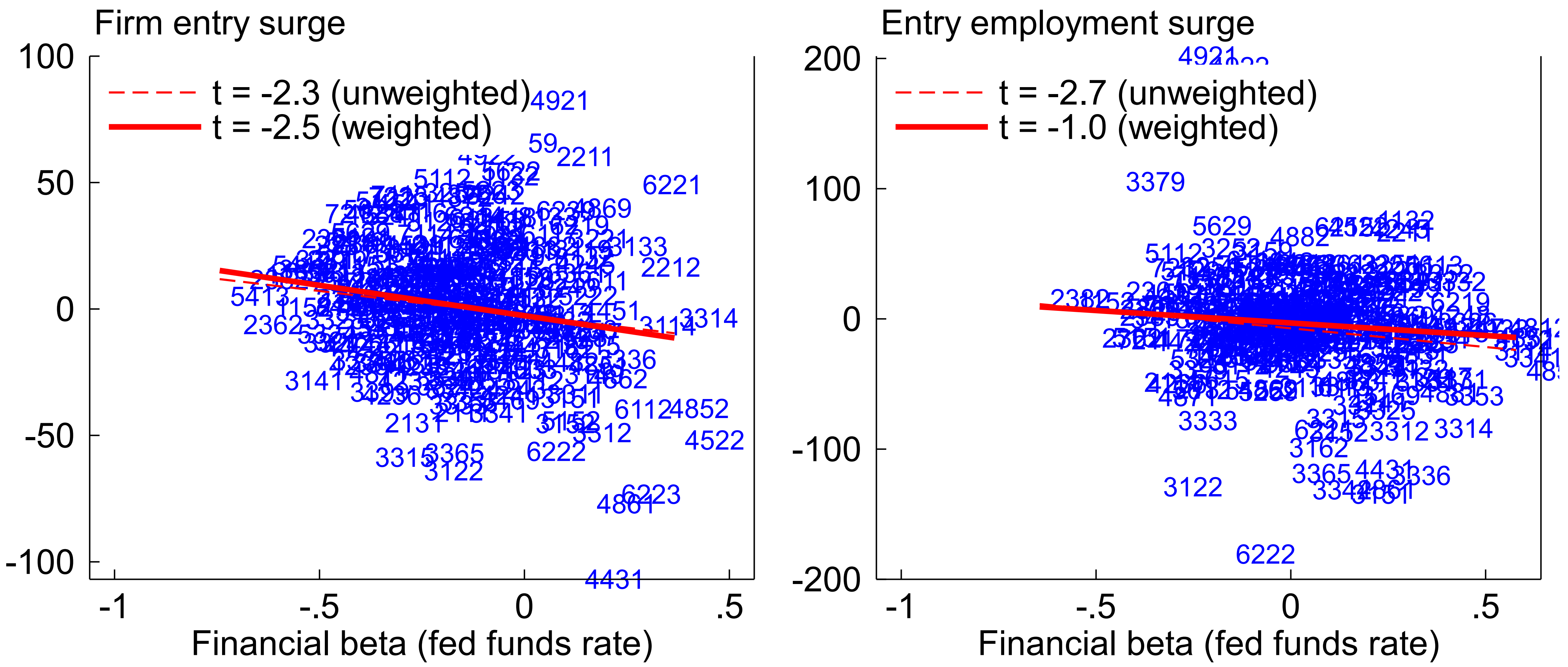
Note: (Log) regulation count from RegData 4.1, 2019.
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, RegData, author calculations.

Financial beta: overall



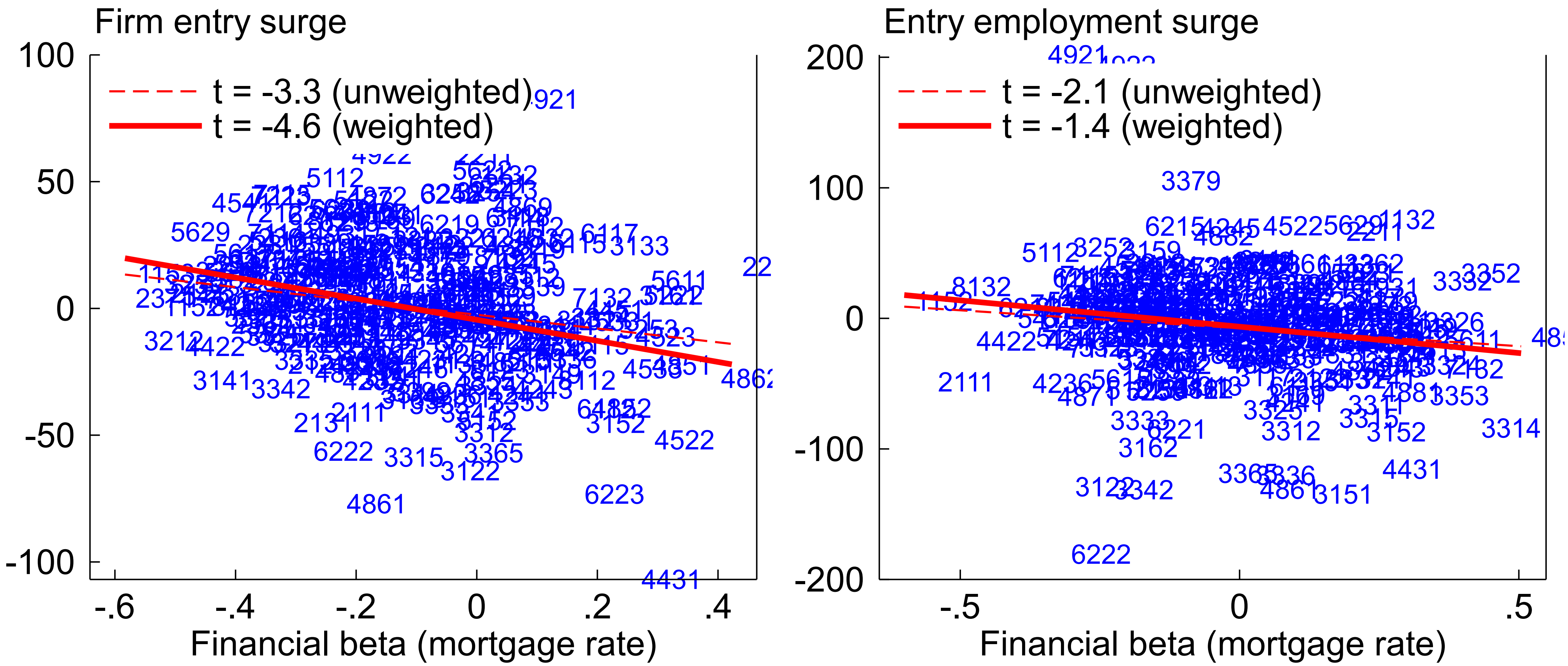
Note: Financial beta is correlation between industry entry and FCI-G, 1991-2019.
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al (2023), author calculations.

Financial beta: Federal funds rate



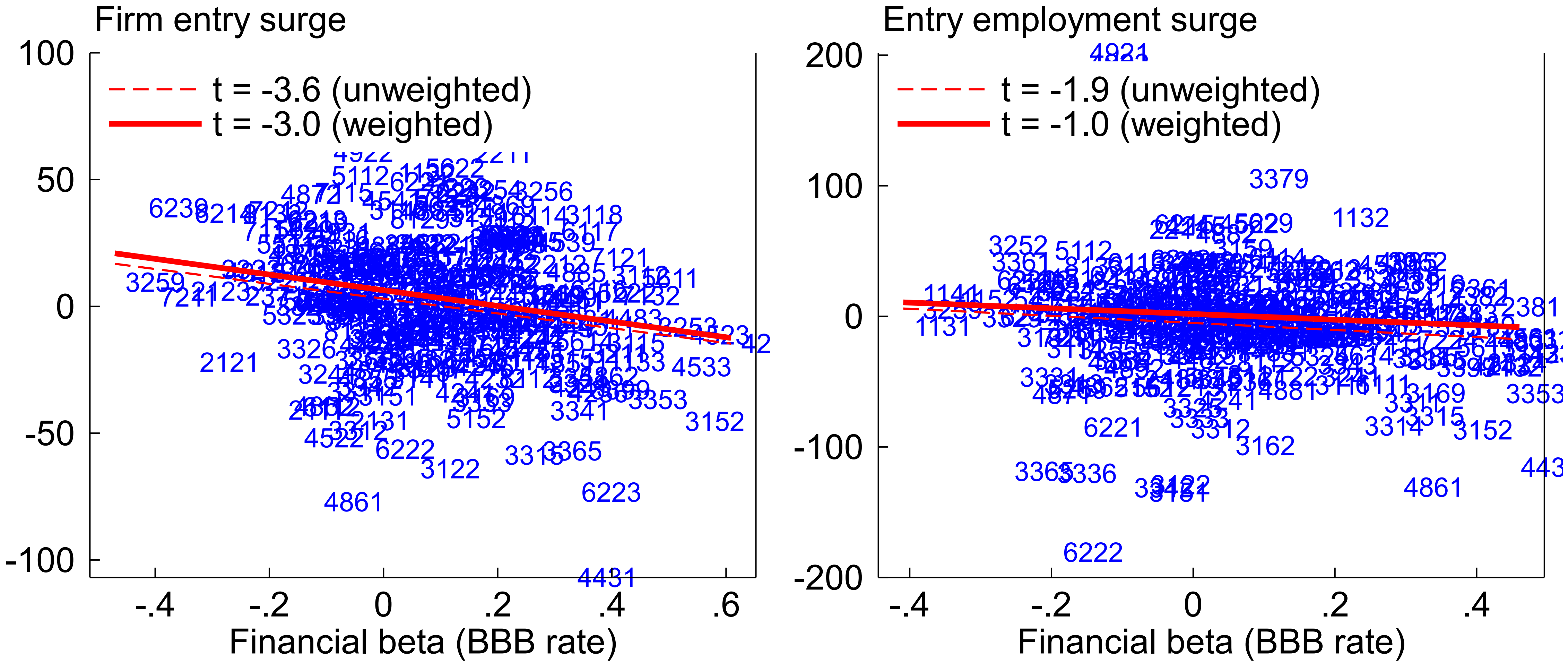
Note: Financial beta is correlation between industry entry and FCI-G FFR component, 1991-2019
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al. (2023), author calculations

Financial beta: 30-year mortgage rate



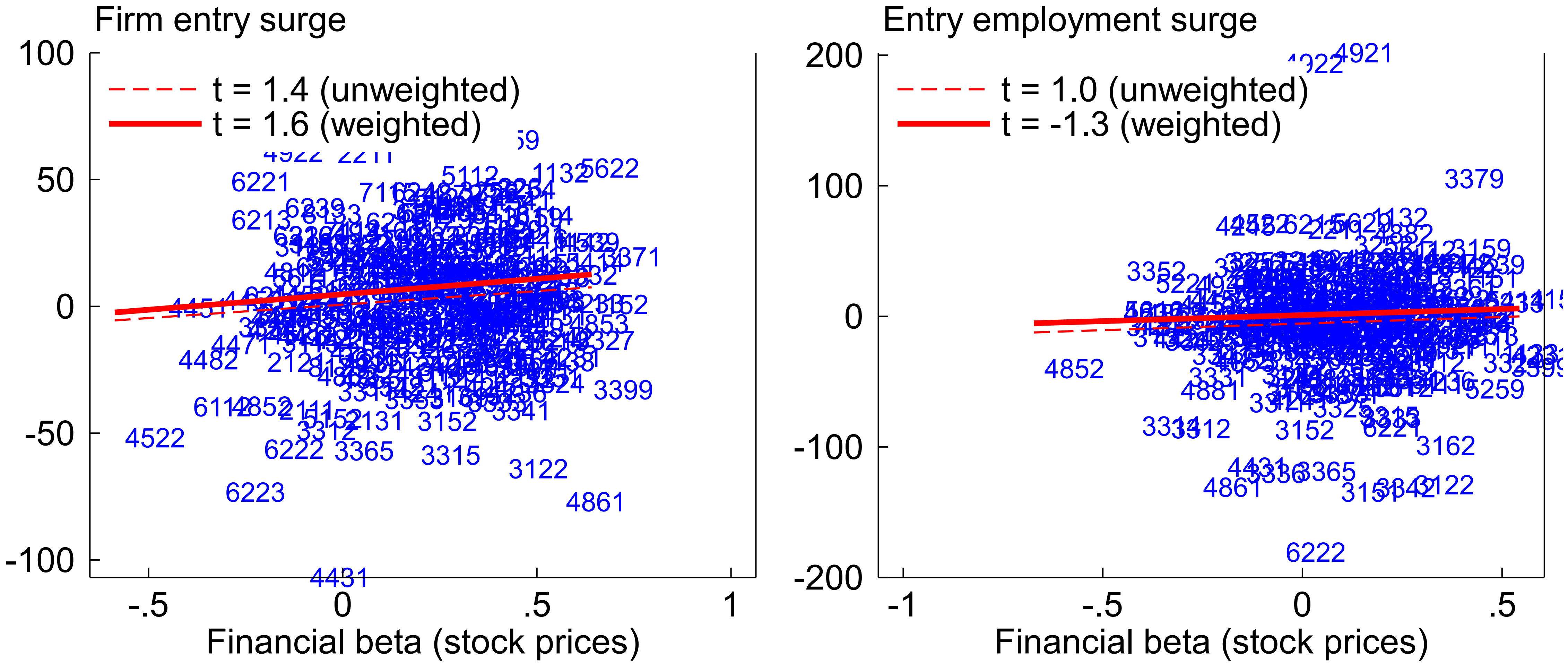
Note: Financial beta is correlation between industry entry and FCI-G 30-yr mortgage rate component, 1991-2019
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al. (2023), author calculations

Financial beta: BBB corporate bond rate



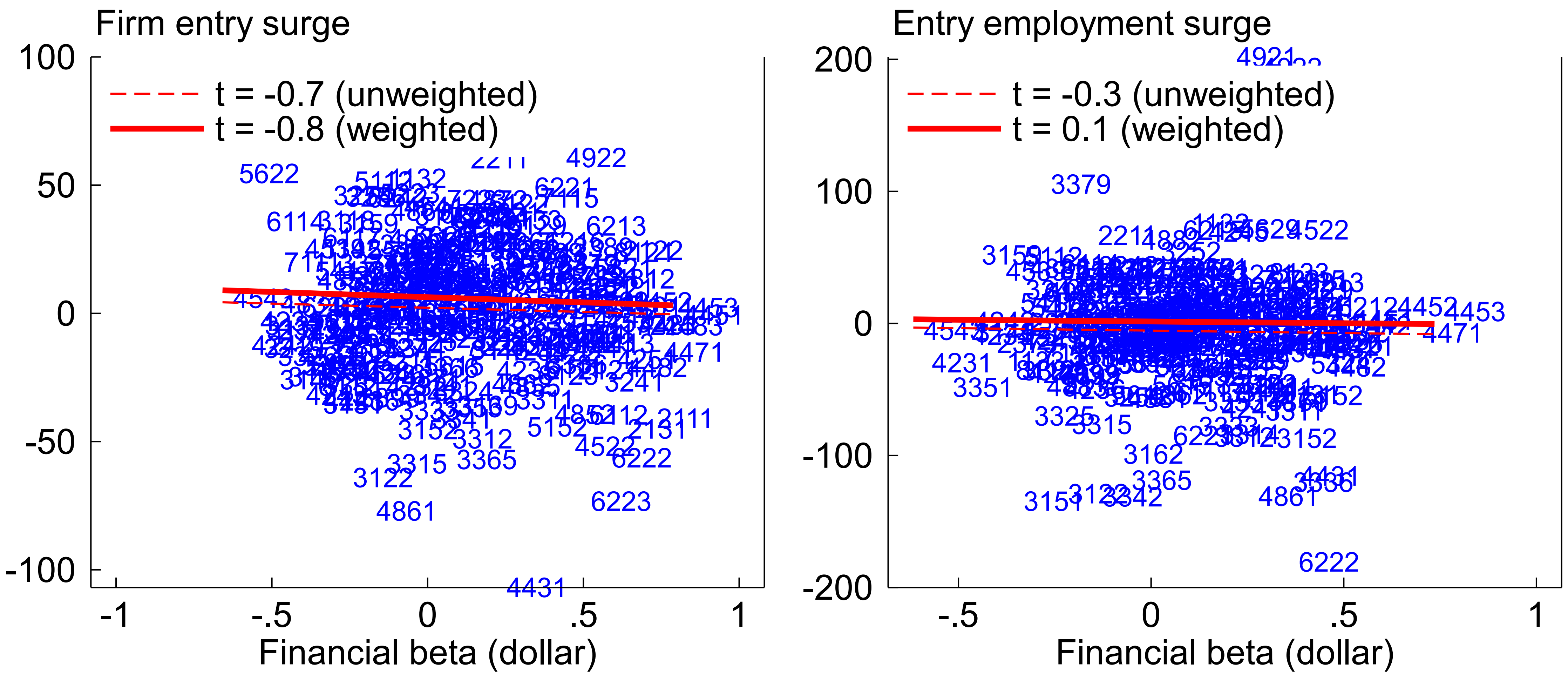
Note: Financial beta is correlation between industry entry and FCI-G corporate BBB rate component, 1991-2019
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (right
Source: BDS, Ajello et al. (2023), author calculations

Financial beta: Stock market value



Note: Financial beta is correlation between industry entry and FCI-G stock price component, 1991-2019
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (right
Source: BDS, Ajello et al. (2023), author calculations

Financial beta: Broad nominal dollar



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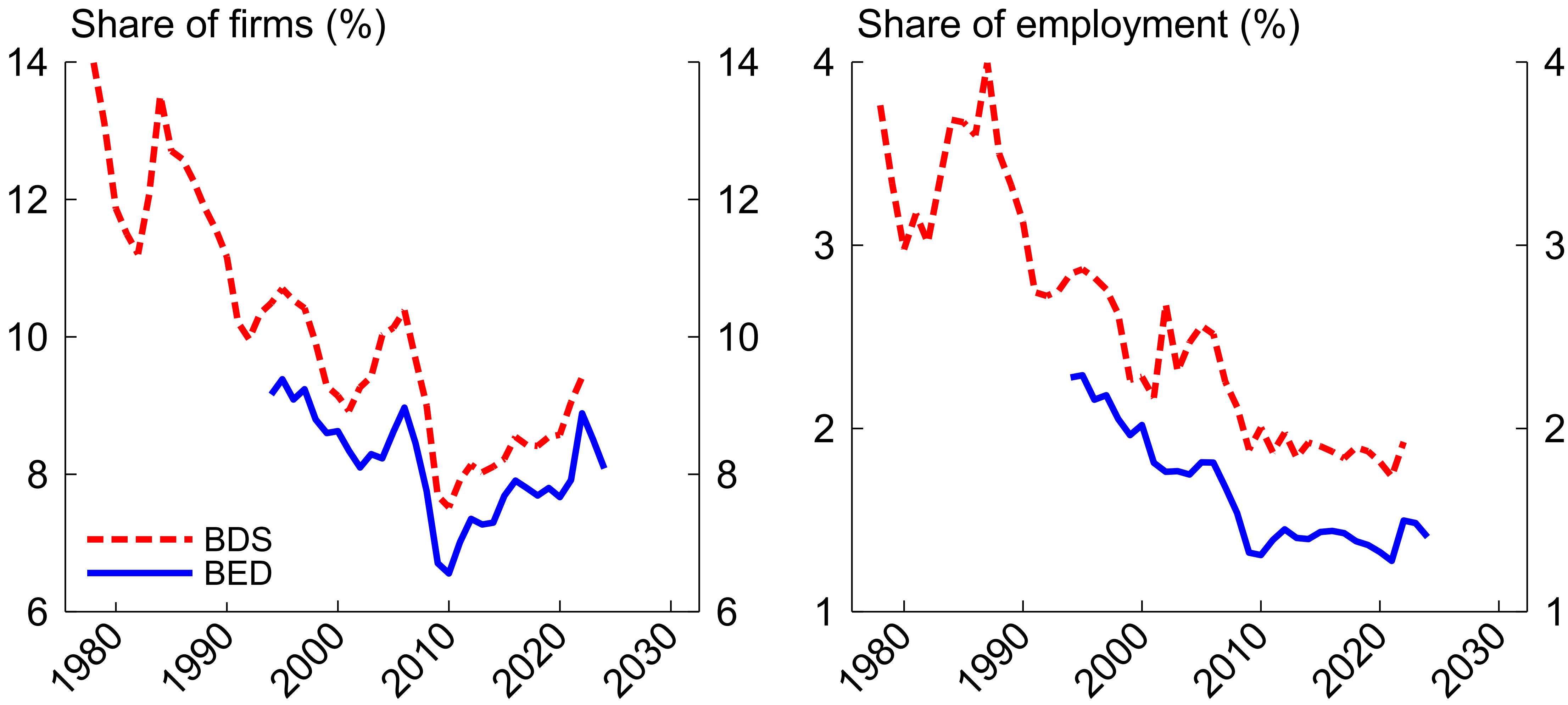
Entry surge in logs, 2021-22 vs. 2013-2019. Regressions weighted by 2013-19 firm count (left) or employment (righ
Source: BDS, Ajello et al. (2023), author calculations

Financial effect variables

	Firm entry	Firm entry	Entry employment	Entry employment
House price	**2.3**	**2.8**	1.1	1.0
Treasury	1.2	1.3	1.7	1.3
Fed funds rate	0.1	**3.3**	-1.2	-0.8
BBB rate	-1.1	1.2	-0.6	-1.0
Mortgage rate	0.6	**-3.0**	1.4	1.3
Stock prices	0.3	1.4	-1.3	0.3
Dollar	-0.2	1.2	-1.3	-0.1
R^2	.08	0.2	.07	.12
Weighted?		Y		Y

Pandemic firm entry surge follows trend decline



Note: Firm entry rates. Right panel uses DHS denominator.
Source: Business Dynamics Statistics (BDS) and Business Employment Dynamics (BED).

Secular decline in business dynamism

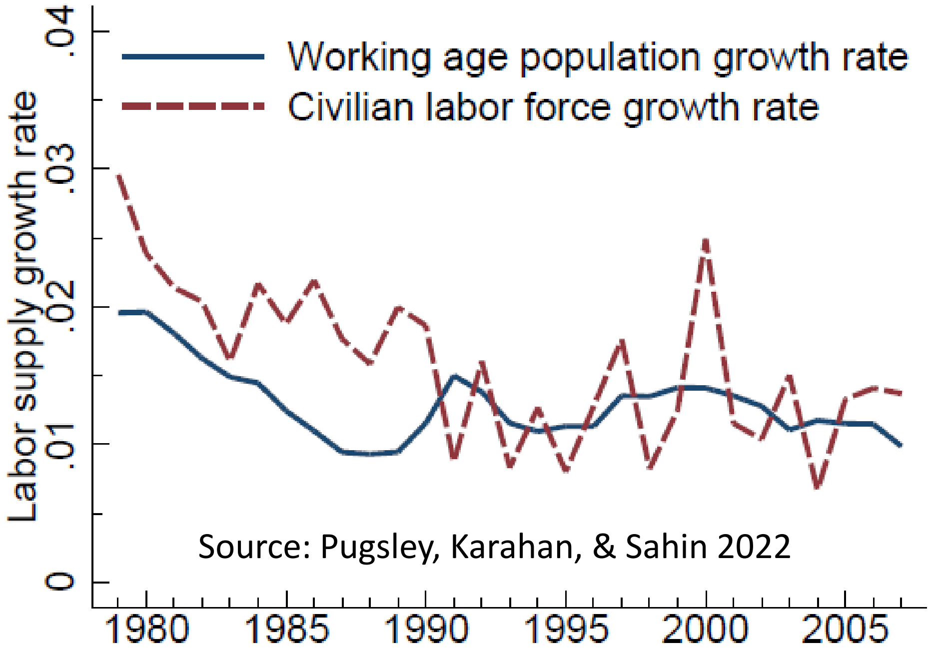
- Large literature on secular decline in business "dynamism" (e.g., Decker et al. 2014)
  - Declining entry rates, job reallocation, worker reallocation, migration
  - Weaker productivity "selection" (correlation between firm/establishment productivity and growth) (Decker et al. 2020)
  - Rising average firm size/concentration
  - Implications for aggregate job creation (Haltiwanger, Jarmin, & Miranda 2013), productivity (Decker et al. 2017, 2020), business cycle (Pugsley & Sahin 2019)

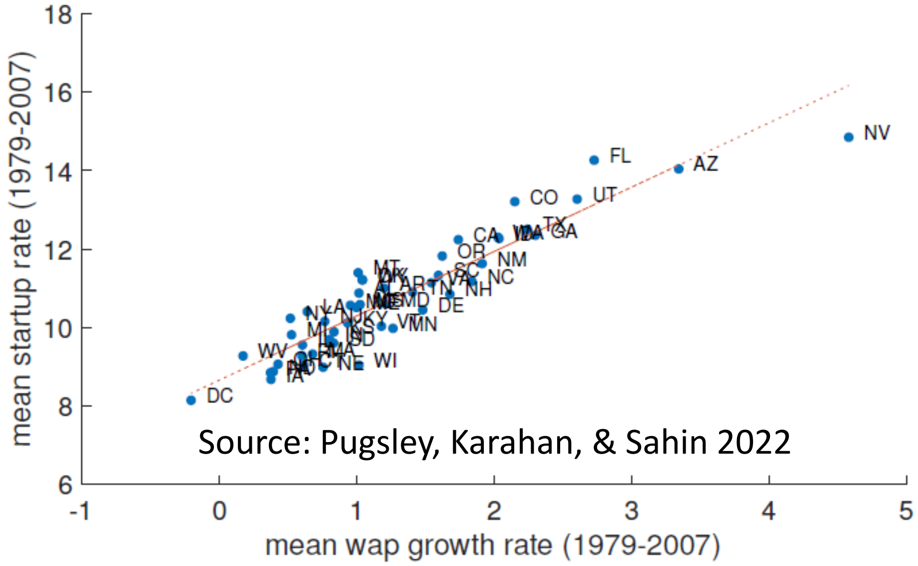
Secular decline in business dynamism

- Causes/consequences explored in literature
  - Demographics (Pugsley, Karahan, & Sahin 2022; Hathaway & Litan 2014; Ozimek 2017)
  - Regulatory/business policy environment (Davis & Haltiwanger 2015; Autor, Kerr, & Kugler 2007; Goldschlag & Tabarrok 2018; Johnson & Kleiner 2020)
  - Change in business model (e.g., retail consolidation, Decker et al. 2016; shift to nonemployers Abraham et al. 2019, Bento & Restuccia 2022)
  - Rising market power (De Loecker, Eeckhout, Mongey 2022; Albrecht & Decker 2024; Foster et al. 2024)
  - Knowledge investment or diffusion (De Ridder 2021, Akcigit & Ates 2023)
  - Debates about “skewness” and whether the decline is real (Guzman & Stern 2020)

Demographics

- In standard models, business entry is facilitated by labor force growth:
  - Slow population growth → Slow labor force growth → less entry (Pugsley, Karahan, & Sahin forthcoming)
  - But note: labor force growth decline concentrated in the 1980s
- Other potential population-related mechanisms: Hathaway & Litan (2014); Ozimek (2017)

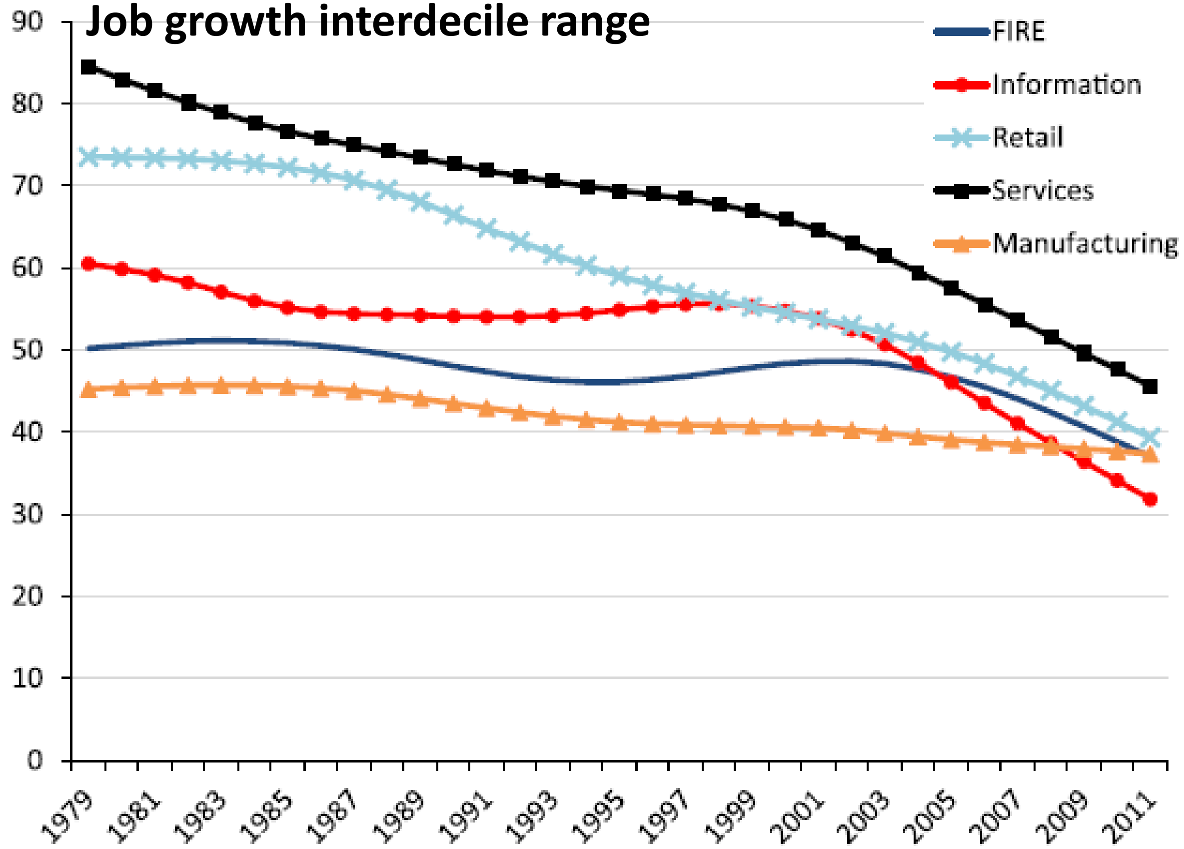


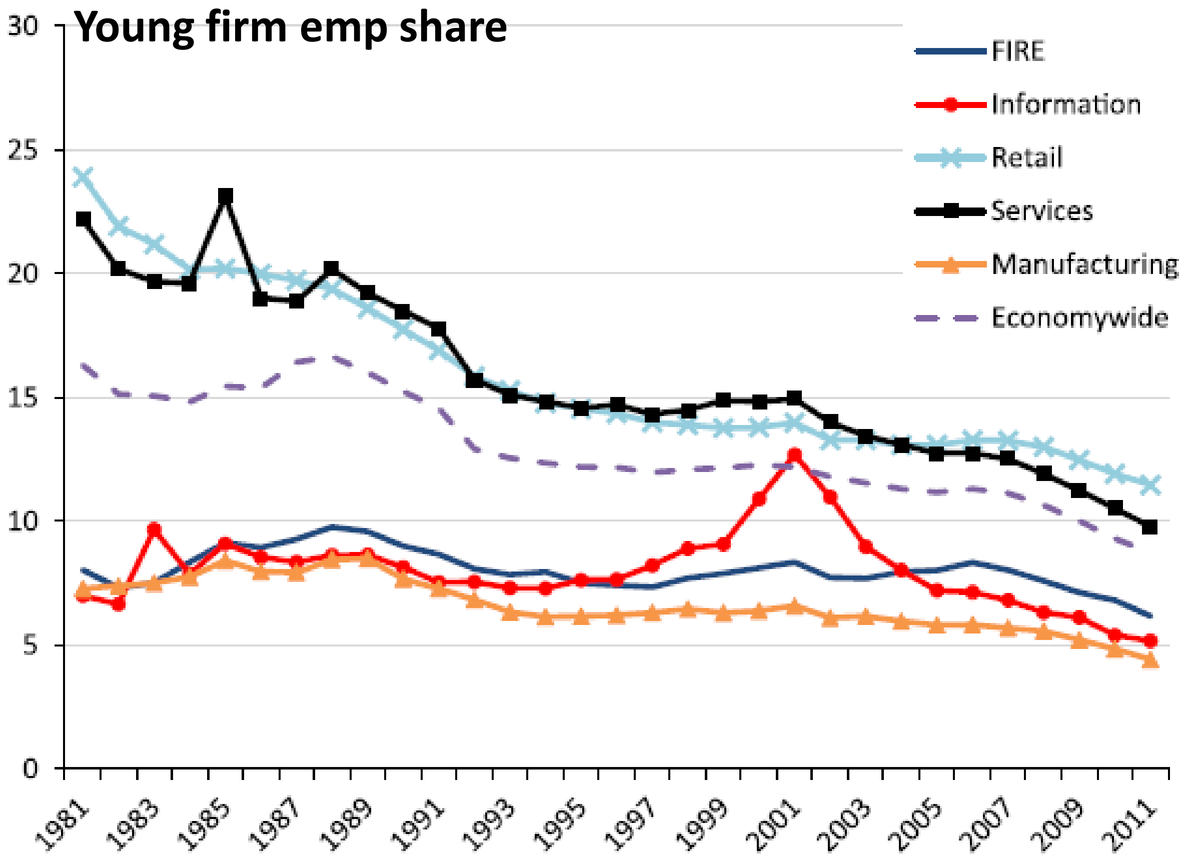


Regulatory environment

- “Death by 1000 cuts” (e.g., Davis & Haltiwanger 2015)
  - Unlawful discharge (Autor, Kerr, & Kugler 2007)
  - Occupational licensing (Johnson & Kleiner 2020)
  - Zoning & other limits on mobility
  - Federal regulation count? No clear relationship with estab. formation (Goldschlag & Tabarrok 2018)

Changing business models

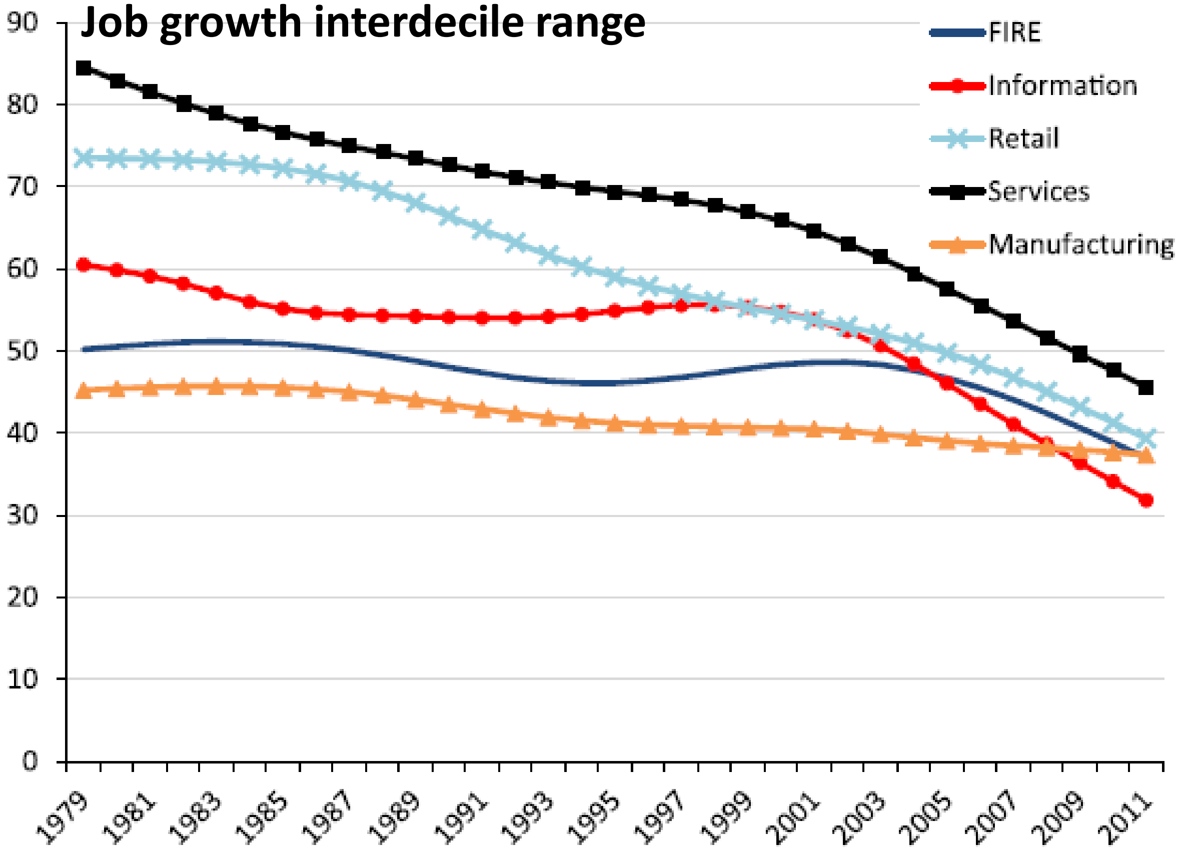


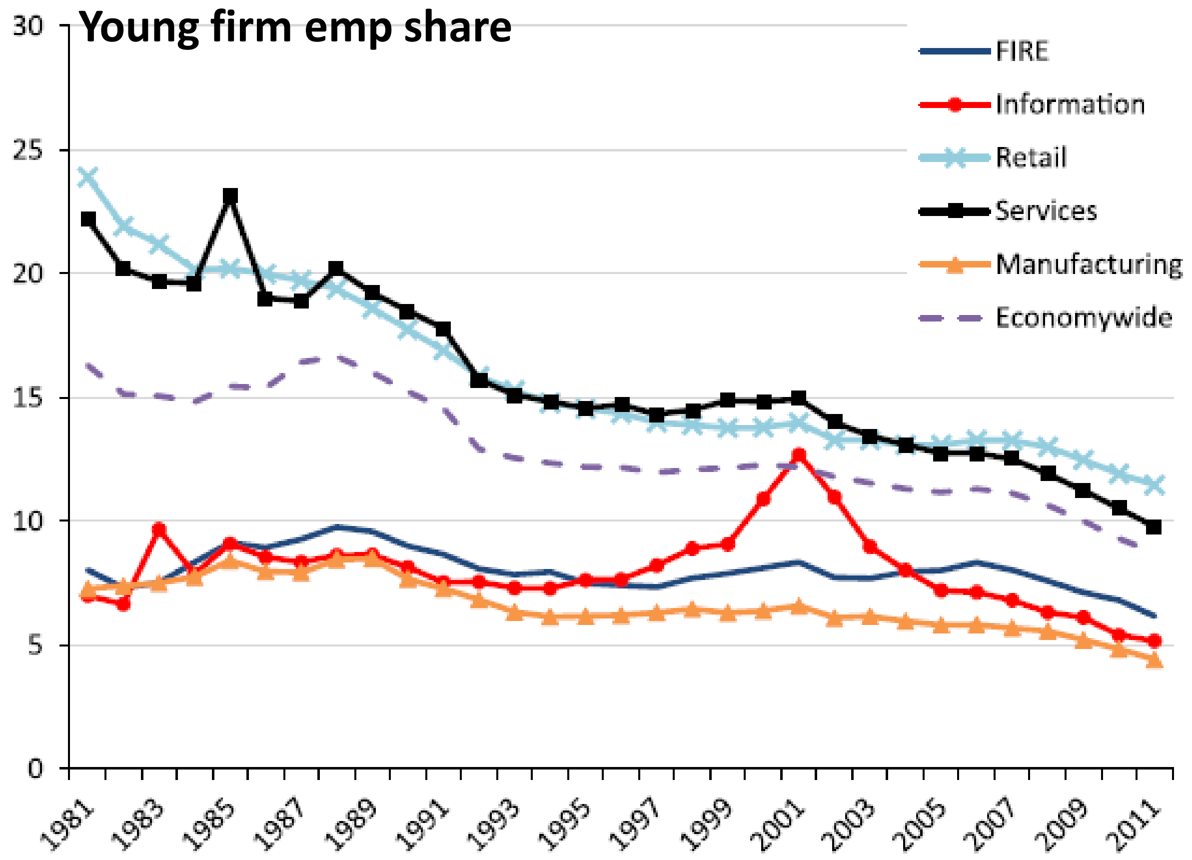


Source: Decker, Haltiwanger, Jarmin, & Miranda (2016) EER

Changing business models

- Retail: decline of "mom and pop" entrepreneurship in favor of "big box" retailers.
  - 1980s-1990s retail consolidation (rise of "big box" retail) was productivity enhancing (Foster et al. 2006, 2016)





Source: Decker, Haltiwanger, Jarmin, & Miranda (2016) EER

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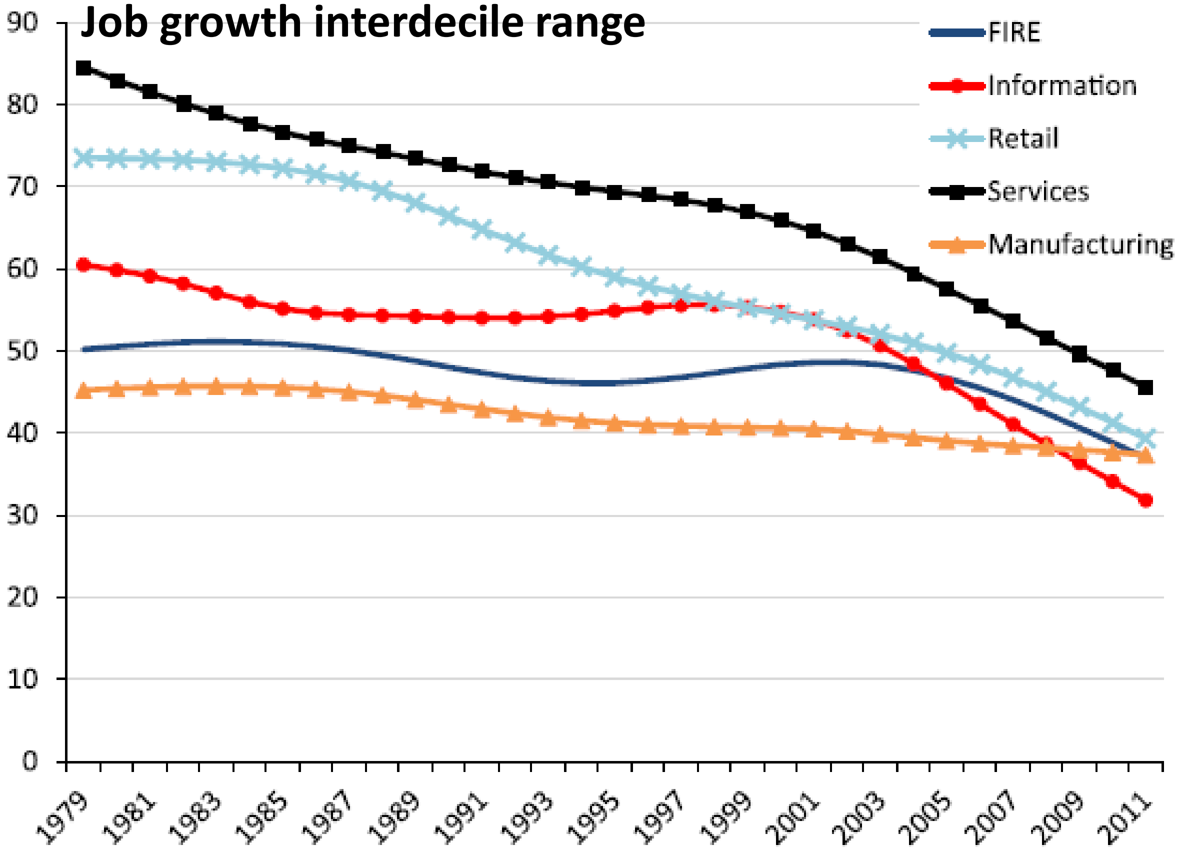
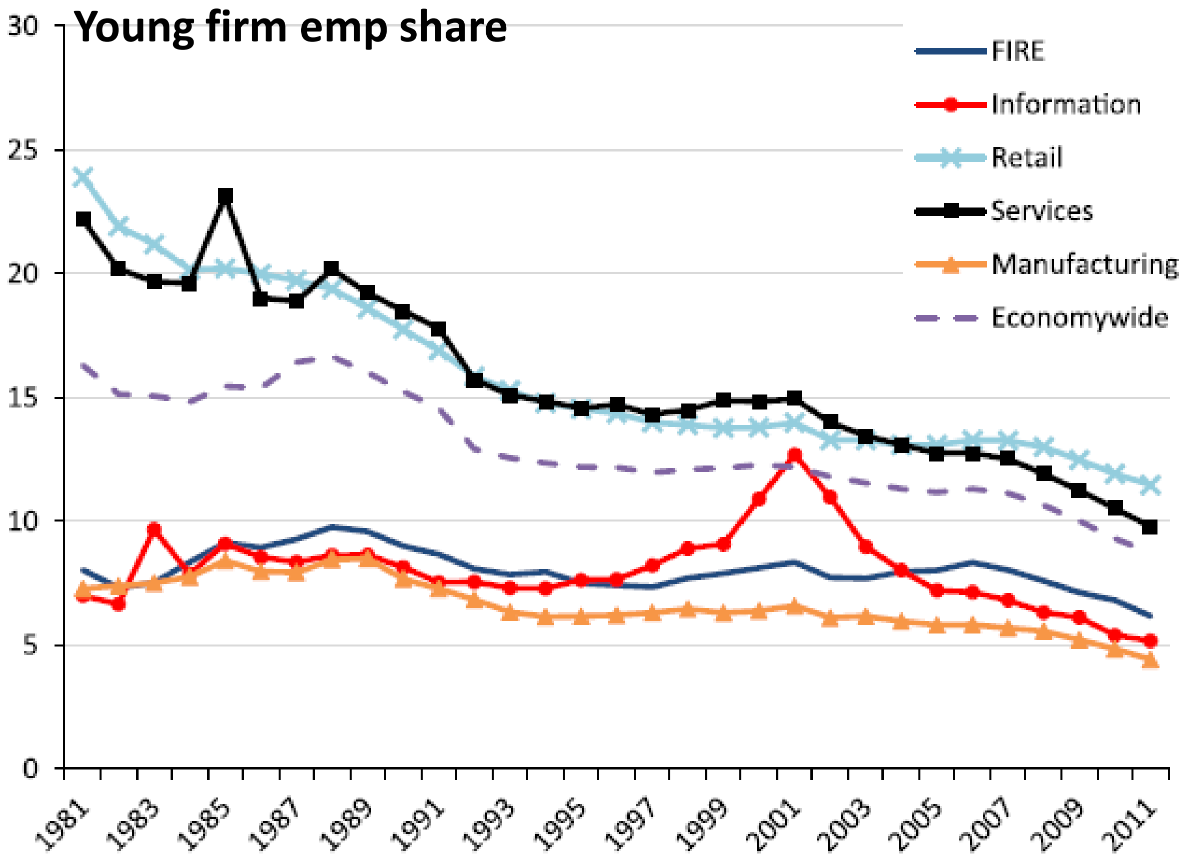


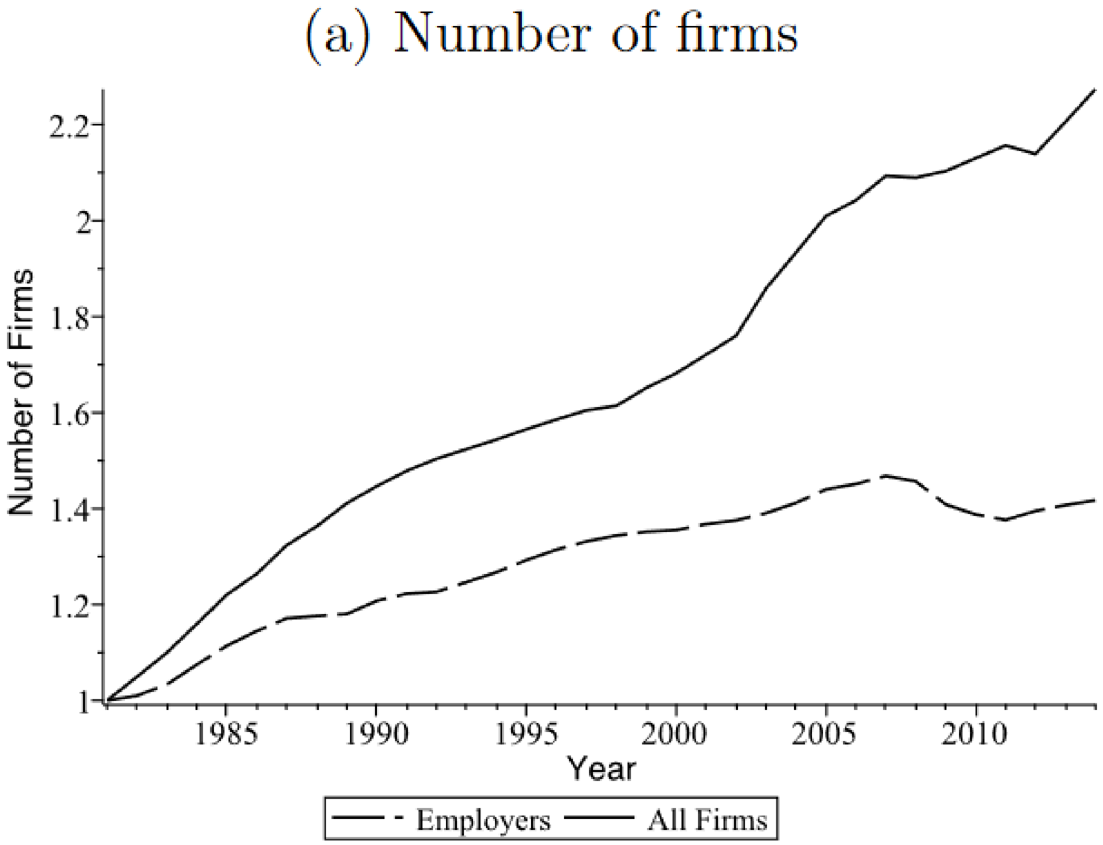
- Tech, information decline starts after ~2000



Source: Decker, Haltiwanger, Jarmin, & Miranda (2016) EER

Changing business models (2)

- Shift to nonemployer entrepreneurship (Bento & Restuccia 2022)
- Rise of "gig" economy?
  - Perhaps limited to transportation sector (Abraham et al. 2019)



Market power

- Rising market power/monopolies (De Loecker, Eeckhout, & Mongey 2022)
  - Market power makes firms less responsive to shocks (↓ reallocation), deters entry

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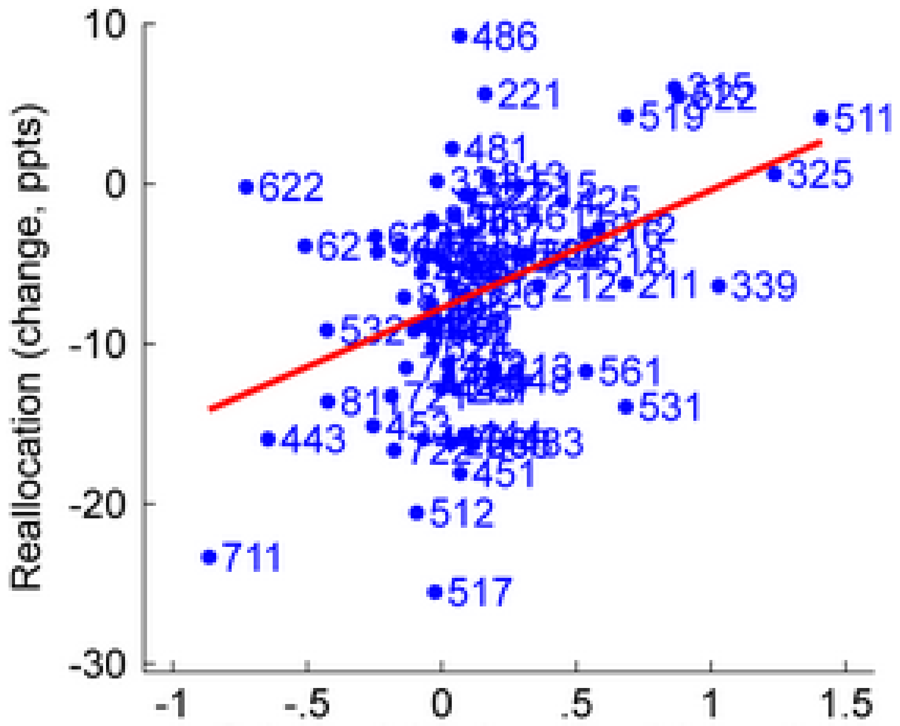
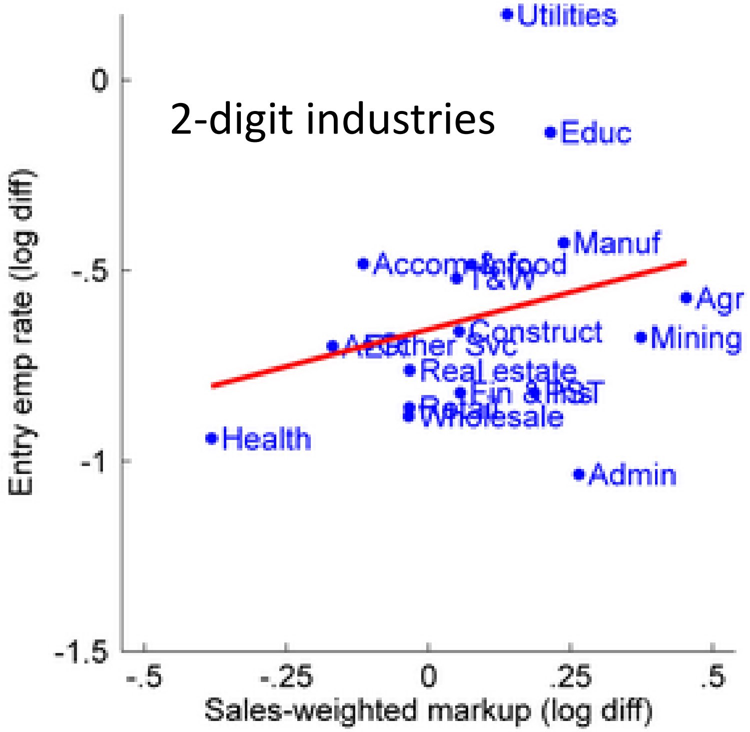
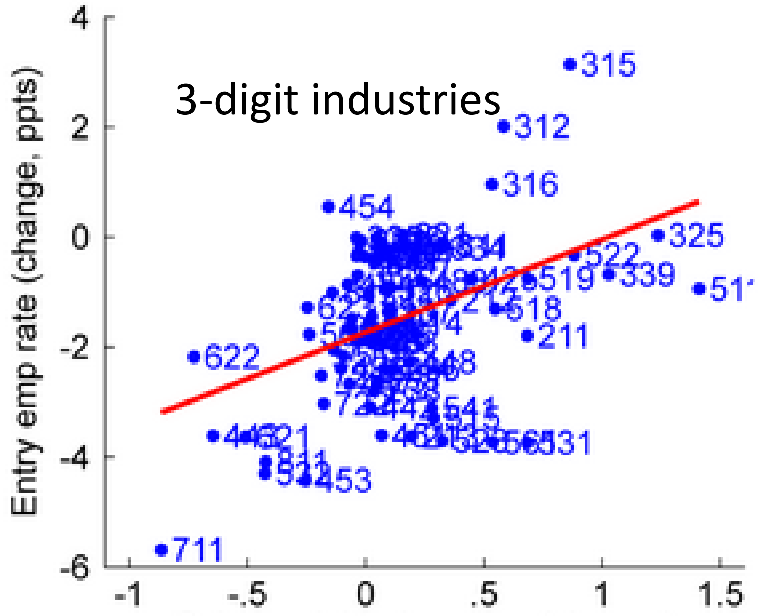






Source: Albrecht & Decker 2024

Knowledge investment or diffusion

- Higher entry costs due to rising importance of intangible capital (De Ridder 2021)
- Declining pace of knowledge diffusion from superstar firms (Akcigit & Ates 2023; Autor et al. 2020; Andrews, Criscuolo, & Gal 2016)
  - Perhaps more relevant for post-2000 decline of high growth young firms, less relevant in pre-2000 period?

Is the decline real?

- Guzman & Stern (2020): Model for identifying high-potential entrepreneurs at (or shortly after) founding

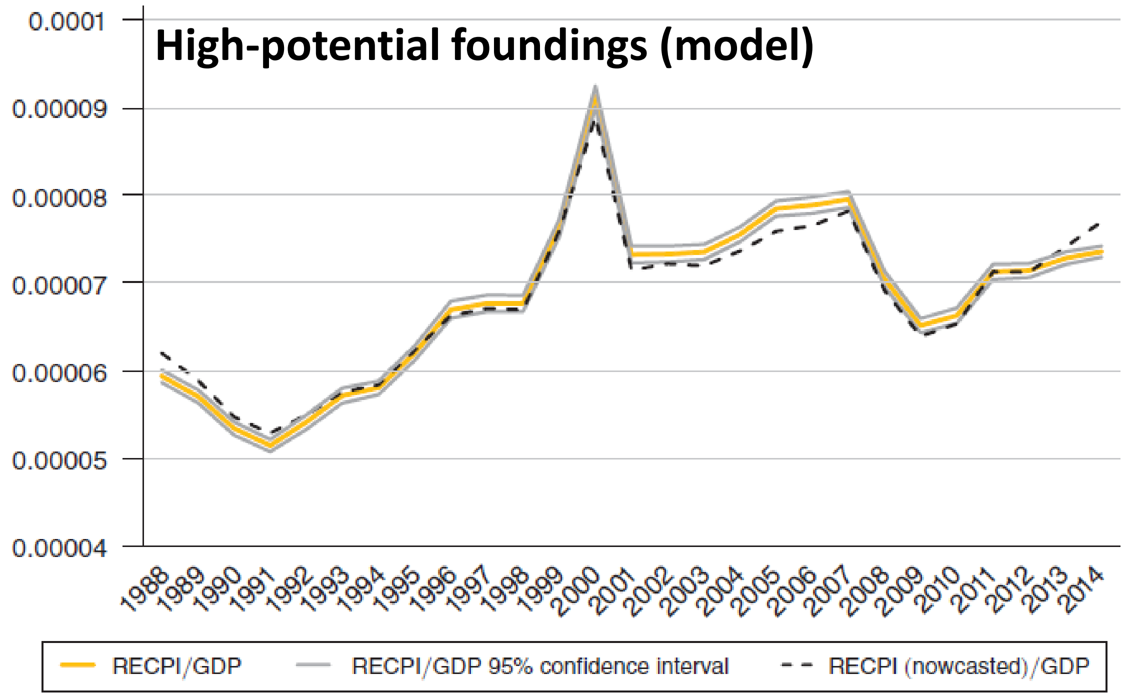
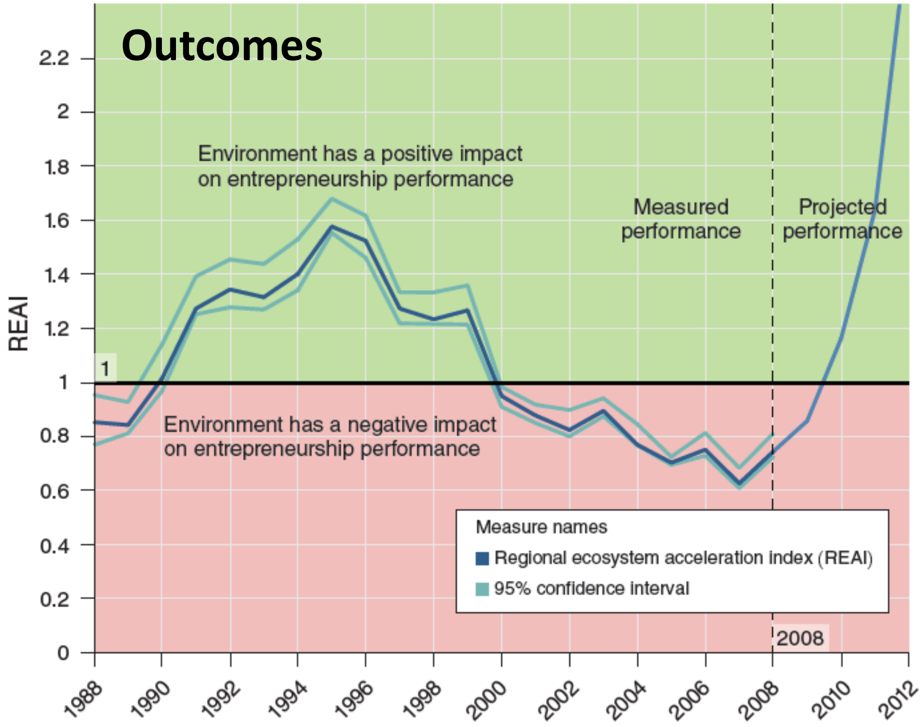


Figure 3. US Aggregate Entrepreneurship Regional Entreprenuership Cohort Potential Index (RECPI) by Year



- Model says: High-potential foundings still robust after 2000
- But… outcomes lower than model expects
  - Consistent with post-2000 decline in high-growth firms & tech documented elsewhere

Explaining the (pre-pandemic) decline in dynamism

- Demographics (1980s?), regulation likely play some role
- Changing business models
  - Retail consolidation apparent in pre-2000 period—productivity enhancing
  - Shift to nonemployers?
- Market power story matches aggregate time series; less apparent in industry cross section
  - Some debate over markup measurement; e.g. Bond et al. (2021); Foster, Haltiwanger, & Tuttle (2024)
- Slowing knowledge diffusion, rising intangibles—potential stories especially for post-2000 decline of high-growth startups
- High-potential foundings (Guzman & Stern 2020) can still be robust without converting to growth outcomes

There is likely no single explanation for the 40-year dynamism decline.